This is a working notes focusing on deriving the c-recursion relation for 2D $\mathcal{N}=1$ superconformal blocks in arbitrary plumbing channel. Written by AI (an individual), not by AI (artificial intelligence).

1 Virasoro conformal blocks

1.1 c-recursion

We first review the recipe for Virasoro conformal blocks, following [**Cho-Collier-Yin**], with the genus-2 conformal block in the theta channel as an example. The derivation can be generalized to arbitrary genus and arbitrary plumbing channel. In this case, the Virasoro conformal block is defined as

$$\begin{aligned} F\left(h_1,h_2,h_3,c\right) = \sum_{|A|=|B|,|C|=|D|,|E|=|F|} & q_1^{|A|} q_2^{|C|} q_3^{|E|} G_{h_1}^{AB} G_{h_2}^{CD} G_{h_3}^{EF} \\ & \times \rho\left(L_{-A}\nu_1, L_{-C}\nu_2, L_{-E}\nu_3\right) \rho\left(L_{-B}\nu_1, L_{-D}\nu_2, L_{-F}\nu_3\right), \end{aligned} \tag{1.1}$$

where A is a partition of $n=|A|$ in descending order, L_{-A} stands for a string of lowering operators made of L_{-m}'s, G_h^{AB} is the inverse of the Gram matrix $\langle h|L_{-A}^\dagger L_{-B}|h\rangle$, and ρ is the 3 point function of Virasoro descendents of the canonically normalized primaries ν_i with the structure constant stripped off:

$$\langle\langle \xi_3 \otimes \bar{\xi}_3 | \xi_2 \otimes \bar{\xi}_2(z,\bar{z}) | \xi_1 \otimes \bar{\xi}_1 \rangle = C_{321} \rho\left(\xi_3,\xi_2,\xi_1|z\right) \rho\left(\bar{\xi}_3,\bar{\xi}_2,\bar{\xi}_1|\bar{z}\right). \tag{1.2}$$

We will also abbreviate $\rho(\xi_1,\xi_2,\xi_3|1)$ as $\rho(\xi_1,\xi_2,\xi_3)$.

The stretegy of computing this conformal block is to treat it as a meromorphic function in c at fixed h_i. The poles on the complex c-plane can be located at the values where the Gram matrix develops a null state, and the whole conformal block can be determined by its large-c behavior and the residue at these poles. Namely,

$$F(h_1,h_2,h_3,c) = F(h_1,h_2,h_3,c\to\infty) + \sum_{i,r\geq 2,s\geq 1} \frac{W_{rs}^{(i)}(h_1,h_2,h_3,c)}{c-c_{rs}(h_i)}. \tag{1.3}$$

Note that we are considering the case where all the internal weights h_i are different, hence there are no double poles from two internal multiplets admits a null state simutaneously.

The poles at $c=c_{rs}(h_i)$ arises when the i-th intermediate multiplet develops a null state. The Kac formula then tells us

$$c_{rs}(h) = 1+6\left(b_{rs}(h)+b_{rs}(h)^{-1}\right)^2, \quad b_{rs}(h)^2 = \frac{rs-1+2h+\sqrt{(r-s)^2+4(rs-1)h+4h^2}}{1-r^2} \tag{1.4}$$

where $r = 2, 3, 4, \cdots, s = 1, 2, 3, \cdots$. Conversely, let's denote the weight h that develops a null state at level rs at fixed c by $d_{rs}(c)$. At this value of c, the multiplet with lowest weight state $\nu_{d_{rs}}$ admits a null state $\chi_{rs} = \sum_{|M|=rs} \chi_{rs}^M L_{-M} \nu_{d_{rs}}$. In this case, the residue $W_{rs}^{(i)}$ receives only contributions from the descendants of this null state. For example when $i = 1$, one have

$$\begin{aligned} W_{rs}^{(1)} = & -\frac{\partial c_{rs}(h_1)}{\partial h_1} \lim_{h_1 \to d_{r,s}} \frac{h_1 - d_{rs}}{\| \sum_{|M|=rs} \chi_{rs}^M L_{-M} \nu_h \|^2} \\ & \times \sum_{|A|=|B|,|C|=|D|,|E|=|F|} q_1^{|A|+rs} q_2^{|C|} q_3^{|E|} G_{h_1+rs}^{AB} G_{h_2}^{CD} G_{h_3}^{EF} \\ & \times \rho\left(L_{-A}\chi_{rs}, L_{-C}\nu_2, L_{-E}\nu_3\right) \rho\left(L_{-B}\chi_{rs}, L_{-D}\nu_2, L_{-F}\nu_3\right). \end{aligned} \tag{1.5}$$

Using the factorization relation

$$\begin{aligned} \rho\left(L_{-N}\chi_{rs}, L_{-M}\nu_2, L_{-P}\nu_3\right) &= \rho\left(L_{-N}\nu_{d_{rs}+rs}, L_{-M}\nu_2, L_{-P}\nu_3\right) \rho\left(\chi_{rs}, \nu_2, \nu_3\right) \\ \rho\left(L_{-N}\nu_1, L_{-M}\chi_{rs}, L_{-P}\nu_3\right) &= \rho\left(L_{-N}\nu_1, L_{-M}\nu_{d_{rs}+rs}, L_{-P}\nu_3\right) \rho\left(\nu_1, \chi_{rs}, \nu_3\right) \\ \rho\left(L_{-N}\nu_1, L_{-M}\nu_2, L_{-P}\chi_{rs}\right) &= \rho\left(L_{-N}\nu_1, L_{-M}\nu_2, L_{-P}\nu_{d_{rs}+rs}\right) \rho\left(\nu_1, \nu_2, \chi_{rs}\right) \\ \rho\left(L_{-N}\chi_{rs}, L_{-M}\nu_2, L_{-P}\chi_{rs}\right) &= \rho\left(L_{-N}\nu_{d_{rs}+rs}, L_{-M}\nu_2, L_{-P}\nu_{d_{rs}+rs}\right) \rho\left(\chi_{rs}, \nu_2, \chi_{rs}\right), \end{aligned} \tag{1.6}$$

one can then arrive at

$$W_{rs}^{(1)} = -\frac{\partial c_{rs}(h_1)}{\partial h_1} A_{rs}^c q_1^{rs} [P_c^{rs}(h_3, h_2)]^2 F(h_1 \to h_1 + rs, c \to c_{rs}(h_1)), \tag{1.7}$$

where the quantity A_{rs}^c and the fusion polynomial $P_c^{rs}(h_3, h_2)$ are defined as

$$\begin{aligned} A_{rs}^c &= \lim_{h \to d_{rs}} \frac{h - d_{rs}}{\| \sum_{|M|=rs} \chi_{rs}^M L_{-M} \nu_h \|^2} = \frac{1}{2} \prod_{m=1-r}^{r} \prod_{n=1-s}^{s} \left(m b_{rs} + n b_{rs}^{-1}\right)^{-1}, \quad (m,n) \neq (0,0), (r,s) \\ P_c^{rs}(h_1, h_2) &\equiv \rho\left(\chi_{rs}, \nu_2, \nu_1\right) = \prod_{p=1-r \text{ step } 2}^{r-1} \prod_{q=1-s \text{ step } 2}^{s-1} \frac{\lambda_1 + \lambda_2 + p b_{rs} + q b_{rs}^{-1}}{2} \frac{\lambda_1 - \lambda_2 + p b_{rs} + q b_{rs}^{-1}}{2}, \end{aligned} \tag{1.8}$$

where the products for the fusion polynomial are taken over $p + r = 1 (\text{mod } 2)$ and $q + s = 1 (\text{mod } 2)$, and λ_i are defined by $h_i = \frac{1}{4}(b_{rs} + b_{rs}^{-1})^2 - \frac{1}{4}\lambda_i^2$. The other residues $W_{rs}^{(2)}, W_{rs}^{(3)}$ follows similarly.

The large-c limit $F(h_1, h_2, h_3, c \to \infty)$ of the conformal block can be computed by the following. Denoting by $\langle A \rangle$ the number of non-L_{-1} generators in L_{-A}. The key observation is that, the Gram matrix element G_{AB} scales like $c^{\langle A \rangle}$ for $A = B$, no faster than $c^{\langle A \rangle - 1}$ for $\langle A \rangle = \langle B \rangle$ but $A \neq B$, and no faster than $c^{\min(\langle A \rangle, \langle B \rangle)}$ for $\langle A \rangle \neq \langle B \rangle$. This can be directly seen from the Virasoro algebra, where the leading term in c is given by the commutators $[L_m, L_{-m}]$. One can then start from the basis of states $\{L_{-A}\nu, |A| = n\}$ at level n, apply Gram-Schmidt orthogonalization, and obtain a basis of orthogonal states $\mathcal{L}_{-A}\nu$. These states are of the form

$$\mathcal{L}_{-A}\nu = L_{-A}\nu + \sum_{|B|=n, \langle B \rangle \leq \langle A \rangle, B \neq A} g_{AB}(c, h) L_{-B}\nu, \tag{1.9}$$

where $g_{AB}(c,h)$ is order $\mathcal{O}(c^0)$ for $B < A$ and $\mathcal{O}(c^{-1})$ for $\langle B\rangle = \langle A\rangle$. In this basis, the conformal block can be written as

$$F(h_1,h_2,h_3,c) = \sum_{A,B,C} q_1^{|A|} q_2^{|B|} q_3^{|C|} \frac{\rho(\mathcal{L}_{-A}\nu_1, \mathcal{L}_{-B}\nu_2, \mathcal{L}_{-C}\nu_3)^2}{\|\mathcal{L}_{-A}\nu_1\|^2 \|\mathcal{L}_{-B}\nu_2\|^2 \|\mathcal{L}_{-C}\nu_3\|^2} \tag{1.10}$$

where $\|\mathcal{L}_{-A}\nu_1\|^2$ is of order $c^{\langle A\rangle}$.

Now let's consider the scaling behavior of $\rho(\mathcal{L}_{-A}\nu_1, \mathcal{L}_{-B}\nu_2, \mathcal{L}_{-C}\nu_3)$ in the $c \to \infty$ limit. In this limit, only the term of order $c^{(\langle A\rangle + \langle B\rangle + \langle C\rangle)/2}$ survives in the conformal block. For a general 3-point function $\rho(L_{-A}\nu_1, L_{-B}\nu_2, L_{-C}\nu_3)$, the leading term must come from pairwise contractions $[L_m, L_{-m}]$ of non-L_{-1} Virasoro generators, and the maximum number of such contractions is precisely $(\langle A\rangle + \langle B\rangle + \langle C\rangle)/2$, which gives a scaling behavior of $c^{(\langle A\rangle + \langle B\rangle + \langle C\rangle)/2}$. In each term of $\mathcal{L}_{-A}\nu$, only the $L_{-A}\nu$ term can give this scaling, since $g_{AB}(c,h)$ is subleading for $\langle B\rangle = \langle A\rangle$ and the correlator is subleading for $\langle B\rangle < \langle A\rangle$. Moreover, since only the central term in the commutator contributes to this limit, we have

$$\rho(\mathcal{L}_{-A}\nu_1, \mathcal{L}_{-B}\nu_2, \mathcal{L}_{-C}\nu_3) \to \rho(L_{-A'}\mathbf{1}, L_{-B'}\mathbf{1}, L_{-C'}\mathbf{1})\rho(L_{-1}^{k_A}\nu_1, L_{-1}^{k_B}\nu_2, L_{-1}^{k_C}\nu_3), \tag{1.11}$$

where $\mathbf{1}$ is the identity operator, and we have written $A = A' \cup \{1\}^{k_A}$. Similarly, the 2-point function also admits a similar factorization relation at large c, namely

$$\|\mathcal{L}_{-A}\nu\|^2 \to \|L_{-A'}\mathbf{1}\|^2 \|L_{-1}^{k_A}\nu\|^2. \tag{1.12}$$

Plugging these relations back to the conformal block, one can then arrive at the large-c limit

$$F(h_1,h_2,h_3,c\to\infty) = F(0,0,0,c\to\infty) \times F_{\mathfrak{sl}(2)}(h_1,h_2,h_3), \tag{1.13}$$

where the first factor is the large-c limit of the vacuum block, and the second factor is the global $\mathfrak{sl}(2)$ block that's independent of c. The global $\mathfrak{sl}(2)$ blocks are generally simple and each term in the q-sum can be computed in terms of Pochhammer symbols (See (4.19)-(4.22) of [**Cho-Collier-Yin**]), while the vacuum block in the Schottky frame has a simple product formula (See (5.5)).

Assembling all the pieces, we arrive at the final form for the c-recursion relation for genus-2 Virasoro conformal blocks in the theta channel:

$$\begin{aligned} F(h_1,h_2,h_3,c) = {} & F(0,0,0,c\to\infty) \times F_{\mathfrak{sl}(2)}(h_1,h_2,h_3) \\ & + \sum_{r\geq 2, s\geq 1} -\left(\frac{\partial c_{rs}(h_1)}{\partial h_1}\right) \frac{A^c_{rs} q_1^{rs} [P_c^{rs}(h_3,h_2)]^2}{c - c_{r,s}(h_1)} F(h_1 \to h_1 + rs, c \to c_{rs}(h_1)) \\ & + 2 \text{ permutation in } h_1, h_2, h_3. \end{aligned} \tag{1.14}$$

Using this relation, one can recursively evalute the conformal block $F(h_1,h_2,h_3,c)$ as a q-expansion, as the polar terms are all higher-order in q due to the factor q^{rs}.

1.2 h-recursion

We now review the derivation of h recursion relation for the tours n point block in the necklace channel. Consider the tours with complex structrue τ, and the flat coordinate on the torus has $z \sim z + 2\pi \sim z + 2\pi\tau$. We consider N primary operators inserted at positions $z = w_i$ for $i = 1, \cdots, N$. We set $w_N = 0$ by convention, and write $w_i = 2\pi(\tau - \sum_{k=1}^{i} \tau_k)$. In the necklace channel, the torus is decomposed into N cylinders, of moduli $\tau_1, \tau_2, \cdots, \tau_N$, with $\sum_{k=1}^{N} \tau_k = \tau$. We will also write $q_i = e^{2\pi i \tau_i}$. The torus N-point block is defined as

$$\begin{aligned} &F\left(q_1, h_1, d_1, ..., q_N, h_N, d_N, c\right) \\ &= \sum_{n_1,\dots,n_N=0}^{\infty} \left(\prod_{i=1}^{N} q_i^{n_i}\right) \sum_{|A_1|=|B_1|=n_1} \cdots \sum_{|A_N|=|B_N|=n_N} \left[\prod_{i=1}^{N} \left(G_{h_i}\right)^{A_i B_i} \rho(L_{-B_i}\nu_i, \mu_i, L_{-A_{i+1}}\nu_{i+1})\right]. \end{aligned} \tag{1.15}$$

where h_i denotes the weight of the intermediate primaries ν_i and d_i denote the conformal weight of external primaries μ_i.

The idea of the h recursion relation is to analyze the analytic structure of the conformal block as a meromorphic function in h_i, sharing the same spirit as the c recursion. The gram matrix would be degenerate at $h_i = h_{r_i,s_i}(c)$, where a singular vector $\chi_{r,s}$ with weight $h_{r,s} + rs$ is developed. For a generic c, the matrix element of the inverse gram matrix will develop a simple pole. Therefore, one can make the following ansatz on the Virasoro block

$$F(q_i, h_i, d_i; c) = F_{\text{reg}}(q_i, h_i, d_i; c) + \sum_{i=1}^{N} \sum_{r_i \geq 1, s_i \geq 1} \frac{B_{r,s}^{(i)}}{h_i - h_{r,s}(c)}. \tag{1.16}$$

where F^{reg} has no pole on the entire h_i plane. The rest of the work will be devoted to pinning down the unknown factor $B_{r,s}^{(i)}$ and $F^{\text{reg}}(q_i, h_i, d_i; c)$.

The factor $B_{r,s}^{(i)}$ can be extracted by considering the limit

$$B_{r,s}^{(i)} = \lim_{h_i \to h_{r,s}} \left(h_i - h_{r,s}(c)\right) F(q_i, h_i, d_i; c). \tag{1.17}$$

Under this limit, only the singular vector $\chi_{r,s}$ inside the Verma module of ν_i, as well as the descendant of $\chi_{r,s}$ will contribute to the Virasoro block. Following from the same derivation in the c-recursion, we have

$$B_{r,s}^{(i)}(h_1, \dots \hat{h}_i, h_N) = A_{r,s}^{c} q_i^{rs} P_c^{rs}(h_{i-1}, d_{i-1}) P_c^{rs}(h_{i+1}, d_i) F(h_i \to h_{r,s} + rs, c), \tag{1.18}$$

Now let us try to extract the regular part F_{reg}. The key step is to change the variable from $\{h_i | i = 1, \dots N\}$ to $\{h_1, a_i = h_i - h_1 | i = 2, \dots N\}$, with $a_1 = 0$ as the bookkeeping notation. Then we have

$$F(q_i, h_i, d_i; c) = F_{\text{reg}}(h_1, a_2 \dots, a_N) + \sum_{i=1}^{N} \sum_{r_i \geq 1, s_i \geq 1} \frac{B_{r,s}^{(i)}}{h_1 + a_i - h_{r,s}(c)}. \tag{1.19}$$

In this new variable, $B^{(i)}_{r,s}(h_1,\dots \hat{h}_i, h_N)$ can be viewed as the limit

$$B^{(i)}_{r,s} = \lim_{h_1\to -a_i+h_{r,s}} \left(h_1 + a_i - h_{r,s}(c)\right) F. \tag{1.20}$$

It can thus be treated as a function of $a_2 \dots a_N$ and independent of h_1. It allows us to extract F^{reg} by analyzing the large h_1 behavior of Virasoro block with a_i keep fixed, where the singular term all get suppressed. As we will show, the strict large h_1 limit for the Virasoro block exist, and therefore $F^{\text{reg}}(h_1, a_2 \dots, a_N)$ has no pole at $h_1 \to \infty$. By the Liouville theorem of analytic function, the regular part must be independent of h_1. Therefore, the regular part can be fully determined by the limit

$$F_{\text{reg}}(q_i, a_i, d_i; c) = \lim_{h_1\to\infty} F(q_i, h_i, d_i; c). \tag{1.21}$$

Now we will present the analysis for the large h behavior of the Virasoro block, and compute the regular part F_{reg} explicitly.

To do the power counting for h_1 in the conformal block, it is convenient to introduce an orthogonal basis defined as follows. Let $A = (a_1, \dots, a_n)$ be a partition, with $a_1 \geq \dots \geq a_n \geq 1$. We write

$$[A] = n, \qquad |A| = \sum_{i=1}^{n} a_i, \tag{1.22}$$

for the number of rows and the size of the partition, respectively. Introduce a total order on the set of partitions by declaring $A < B$ if either

1. $[A] < [B]$; or
2. $[A] = [B]$ and $a_j < b_j$ at the first index j for which $a_j \neq b_j$.

For generic c and h, an orthogonal basis of the Verma module can be constructed by applying the Gram–Schmidt procedure to the states $L_{-A}|h\rangle$ at each fixed level, in increasing order with respect to the ordering above. The resulting states take the form

$$\ell_{-A}|h\rangle = L_{-A}|h\rangle + \sum_{\substack{|B|=|A|\\ B<A}} k^A_B(c,h)\, L_{-B}|h\rangle. \tag{1.23}$$

The scaling of $k^A_B(c,h)$ can be determined by the orthogonal condition $\langle h|\ell^\dagger_{-B}\ell_{-A}|h\rangle = 0$ as well as the scaling of the matrix element

$$\langle h|L^\dagger_{-A}L_{-B}|h\rangle = \begin{cases} C_A h^{[A]}\left(1+\mathcal{O}(h^{-1})\right), & A = B,\\ \mathcal{O}\!\left(h^{[A]-1}\right), & [A]=[B], \quad A\neq B,\\ \mathcal{O}\!\left(h^{\min([A],[B])}\right), & [A]\neq[B]. \end{cases} \tag{1.24}$$

One has

$$\begin{aligned} k_B^A(c,h) &\sim \mathcal{O}(h^{-1}), \quad [B]=[A],\ B\neq A; \\ k_B^A(c,h) &\sim \mathcal{O}(h^{0}), \quad [B]<[A]. \end{aligned} \tag{1.25}$$

Therefore, the norm of the basis state $\ell_{-A}|h\rangle$ satisfies

$$\langle h|\ell_{-A}^{\dagger}\ell_{-A}|h\rangle = \langle h|L_{-A}^{\dagger}L_{-A}|h\rangle + \dots . \tag{1.26}$$

where $\dots$ denote the subleading term in h_1. Under the orthogonal basis, the Virasoro block can be represented as

$$F = \sum_{n_1,\dots,n_N=0}^{\infty}\left(\prod_{i=1}^{N} q_i^{n_i}\right)\sum_{|A_1|=n_1}\cdots\sum_{|A_N|=n_N}\left[\prod_{i=1}^{N}\frac{\rho(\ell_{-A_i}h_1, d_i, \ell_{-A_{i+1}}h_1)}{\langle h_1|\ell_{-A_i}^{\dagger}\ell_{-A_i}|h_1\rangle}\right]. \tag{1.27}$$

We also need to determine the asymptotic of three point function $\rho(\ell_{-A}\nu_i, \mu_i, \ell_{-B}\nu_{i+1})$. The key observation is that the three point function $\rho(L_{-A}\nu_i, \mu_i, L_{-B}\nu_{i+1})$ satisfy the same conformal ward identity as $G_{AB} = \rho(L_{-A}\nu_i, \mathbf{1}, L_{-B}\nu_i)$ when h_1 is large. This is because all the terms in the ward identity that involves the action of Virasoro generators on μ_i will produce the subleading contribution in h_1. Therefore, one can effectively replace μ_i with the identity operator, with the caveat of replacing ν_{i+1} in the matrix element with ν_i to ensure the normalization $\rho(\nu_i, \mathbf{1}, \nu_i) = 1$ make sense.

Combining this argument with the scaling of k_B^A (1.25), we conclude

$$\lim_{h_1\to\infty}\frac{\rho(\ell_{-A}\nu_i, \mu_i, \ell_{-B}\nu_{i+1})}{\langle h|\ell_{-A}^{\dagger}\ell_{-A}|h\rangle} = \lim_{h_1\to\infty}\frac{\rho(L_{-A}\nu_i, \mathbf{1}, L_{-B}\nu_{i+1}) + \dots}{\langle h|L_{-A}^{\dagger}L_{-A}|h\rangle + \dots} = \delta_{AB}. \tag{1.28}$$

Therefore, the large h_1 limit of the Virasoro block exist and collapse into

$$\lim_{h_1\to\infty} F(h_1, a_i, d_i) = \sum_{n=0}^{\infty} p(n)(q_1\dots q_N)^n = \prod_{n=1}^{\infty}\frac{1}{1-(q_1\dots q_N)^n}. \tag{1.29}$$

We can thus identify it as the regular block and conclude the h recursion in the necklace channel as

$$F(q_i, h_i, d_i; c) = \prod_{n=1}^{\infty}\frac{1}{1-(q_1\dots q_N)^n} + \sum_{i=1}^{N}\sum_{r_i\geq 1, s_i\geq 1}\frac{B_{r,s}^{(i)}}{h_i - h_{r,s}(c)}. \tag{1.30}$$

with $B_{r,s}^{(i)}$ defined in (1.18).

1.3 h-recursion for sphere n point block in elliptic parameter

In this section, we are going to explore the h recursion for sphere n point block in the elliptic parameter. The idea is to use the fact that the 4-punctured Riemann sphere S^2 is conformal equivalent to the pillow geometry T^2/Z_2 with flat metric, and write down the recursion relation in the pillow frame. As we will see, it provides an efficient resummation over the puncture position moduli z_i and thus providing a very efficient way of computing sphere n point block.

Conformal mapping to pillow geometry

Use conformal symmetry to fix 4 puncture to be at $(0, z, 1, \infty)$. The conformal map from 4-punctured sphere to pillow geometry can be worked out by identifying the Riemann sphere as the x-plane for the elliptic curve[1]

$$y^2 = x(x-z)(x-1). \tag{1.31}$$

In this presentation, the elliptic curve provides a double cover of the x-plane by looking at two branches for y. The single copy of the Riemann sphere can thus be identified as the quotient $T^2/\mathbb{Z}_2$, with $\mathbb{Z}_2$ acting as $y \to -y$ that identify two branches. Four fix point loci of the $\mathbb{Z}_2$ action is the position of 4 punctures.

Introduce the holomorphic one form on the elliptic curve

$$\omega = \frac{dx}{y} = \frac{dx}{\sqrt{x(x-z)(x-1)}} \tag{1.32}$$

We can choose A cycle to be the loop surrounding $(0, z)$ and B cycle to be the loop surrounding $(z, 1)$. The complex structure of the torus can be computed by

$$\tau = \frac{\int_B \omega}{\int_A \omega} = i\frac{K(1-z)}{K(z)}, \tag{1.33}$$

where

$$K(z) := \frac{1}{\pi}\int_0^1 \frac{d\xi}{\sqrt{\xi(1-\xi)(1-z\xi)}} = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; z\right). \tag{1.34}$$

The inverse map from z to τ is given by

$$z = \frac{\theta_2(0|\mathfrak{q})^4}{\theta_3(0|\mathfrak{q})^4}, \quad K(z) = \theta_3(0|\mathfrak{q})^2, \quad \mathfrak{q} = e^{\pi i \tau}. \tag{1.35}$$

One can construct the Abel-Jacobi map from the elliptic curve to the flat torus $\mathbb{C}/\Gamma$ via

$$w(x) = \frac{1}{\theta_3(\tau)^2}\int_0^x \frac{d\xi}{\sqrt{\xi(\xi-z)(\xi-1)}} \tag{1.36}$$

where the factor $\theta_3(\tau)^2$ is introduced such that $w \cong w + 2\pi w \cong w + 2\pi\tau$.

[1]The genus of the curve can be worked out by treating it as a projective variety inside $\mathbb{CP}^2$ and using the adjunction formula to compute the chern class. Use the fact that the top chern class = eular class, one can compute the eular characteristic of the curve to show that it has $\chi = 0$, namely it has genus 1.

Sphere 4 point block in pillow frame

The sphere 4 point block can be understood as the 4 point correlation function with projection operator for the Verma module $\mathbb{P}_{h,\bar{h}}$ inserted. As $\mathbb{P}_{h,\bar{h}}$ commute with the Virasoro generators, it does not transform under the local conformal transformation.

Now we are going to derive the conformal transformation of sphere conformal block in the pillow geometry. The flat metric on the torus is related with the one on the Riemann sphere as

$$ds^2 = dwd\bar{w} = e^{2\sigma(x,\bar{x})}dxd\bar{x}, \tag{1.37}$$

with

$$e^{2\sigma(x,\bar{x})} = \frac{1}{|\theta_3(\tau)|^4}\frac{1}{|x(x-z)(x-1)|}. \tag{1.38}$$

$$\mathcal{F}_c(h_1,h_2,h_3,h_4,h;z) = \theta_3(0|\tau)^{\frac{c}{2}-4\sum_{i=1}^4 h_i} z^{\frac{c}{24}-h_1-h_2}(1-z)^{\frac{c}{24}-h_2-h_3}\left\langle \psi_{43}\left|\mathbb{P}_h\, \mathfrak{q}^{L_0-\frac{c}{24}}\right|\psi_{12}\right\rangle. \tag{1.39}$$

h recursion in the pillow frame

The matrix element can be computed by resolving the projection operator $\mathbb{P}_h$

$$\langle\psi_{34}|\,\mathbb{P}_h\mathfrak{q}^{L_0-\frac{c}{24}}\,|\psi_{12}\rangle = \mathfrak{q}^{h-\frac{c}{24}}\sum_N \mathfrak{q}^N\,\langle\psi_{34}|\,L_{-N}\,|h\rangle\,G^{NM}\,\langle h|\,L_{-M}^\dagger\,|\psi_{12}\rangle\,. \tag{1.40}$$

The subtlety here is that $\langle\psi_{34}|h\rangle$ is not normalized. The normalization of this factor can be determined by compare with the defining equation for sphere 4 point block at $z\to 0$ limit to be

$$\langle\psi_{43}|h\rangle\,\langle h|\psi_{12}\rangle = 16^{h-\frac{c}{24}}. \tag{1.41}$$

Therefore, we can subtract this factor and introduce

$$\mathcal{G}(h_1,h_2,h_3,h_4,h;\mathfrak{q}) = (16\mathfrak{q})^{-h+\frac{c}{24}}\,\langle\psi_{34}|\,\mathbb{P}_h\mathfrak{q}^{L_0-\frac{c}{24}}\,|\psi_{12}\rangle = \sum_N \mathfrak{q}^N\rho(\psi_{34},L_{-N}\nu_h)G^{NM}\rho(L_{-M}\nu_h,\psi_{12}) \tag{1.42}$$

where the factor is normalized by $\rho(\psi_{34},\nu_h) = \rho(\nu_h,\psi_{12}) = 1$. In the same spirit as the c recursion relations derived before, we can analyze the pole structure of $\mathcal{G}$ as an analytic function of h. For generic c, it would suggest the following pole structure

$$\mathcal{G}(h_1,h_2,h_3,h_4,h;\mathfrak{q}) = F_{reg} + \sum_{r,s\geq 1}\frac{C_{r,s}}{h-h_{r,s}(c)}. \tag{1.43}$$

The pole factor $C_{r,s}$ can be extracted by the $h \to h_{r,s}(c)$ limit, where only the null states $\chi_{r,s}$ will contribute in this sum. The norm factor for the null state follows from (1.8). We also have the factoralization relation

$$\rho(\psi_{34}, L_{-N}\chi_{r,s}) = \rho(\psi_{34}, \chi_{r,s})\rho(\psi_{34}, L_{-N}\nu_{h_{r,s}+rs}). \tag{1.44}$$

The factor that is not obvious to determine is $\rho(\psi_{34}, \chi_{r,s})$. It can be determined by doing the transforming back to the sphere coordinate and compare with the three point normalization. The result is that it is still given by the fusion polynomial up to an extra 16 factor

$$\rho(\psi_{34}, \chi_{r,s})\rho(\chi_{r,s}, \psi_{12}) = 16^{rs} P_c^{rs}(h_4, h_3) P_c^{rs}(h_2, h_1). \tag{1.45}$$

Therefore, we have

$$C_{r,s} = A_{r,s}^c (16\mathfrak{q})^{rs} P_c^{rs}(h_4, h_3) P_c^{rs}(h_2, h_1) \mathcal{G}(h \to h_{r,s} + rs, c). \tag{1.46}$$

The regular part of the recursion relation can be determined analyzing the large h asymptotic. One can determine

$$F_{reg} = \prod_{n\geq 1}(1-\mathfrak{q}^{2n})^{-1/2}. \tag{1.47}$$

The derivation using the conformal ward identity for the pillow states. Require conformal wald identity in pillow geometry

Therefore, the block $\mathcal{G}$ satisfy the following h recursion relation

$$\mathcal{G}(h_1, h_2, h_3, h_4, h; \mathfrak{q}) = \prod_{n\geq 1}(1-\mathfrak{q}^{2n})^{-1/2} + \sum_{r,s\geq 1} \frac{C_{r,s}}{h - h_{r,s}(c)}. \tag{1.48}$$

with $C_{r,s}$ given in (1.46). It is convenient to strip off the global block part and define the elliptic block as

$$H(h_1, h_2, h_3, h_4, h; \mathfrak{q}) = \prod_{n\geq 1}(1-\mathfrak{q}^{2n})^{1/2} \mathcal{G}(h_1, h_2, h_3, h_4, h; \mathfrak{q}) \tag{1.49}$$

It would satisfy the following h recursion relation

$$H(h_1, h_2, h_3, h_4, h; \mathfrak{q}) = 1 + \sum_{r,s\geq 1} \frac{B_{r,s}}{h - h_{r,s}(c)}. \tag{1.50}$$

where

$$B_{r,s} = A_{r,s}^c (16\mathfrak{q})^{rs} P_c^{rs}(h_4, h_3) P_c^{rs}(h_2, h_1) H(h \to h_{r,s} + rs, c). \tag{1.51}$$

As a result, the computation of the sphere 4 point block follows from

$$\begin{aligned} \mathcal{F}_c(h_1, h_2, h_3, h_4, h; z) = {} & \theta_3(0|\tau)^{\frac{c}{2} - 4\sum_{i=1}^4 h_i} z^{\frac{c}{24} - h_1 - h_2} (1-z)^{\frac{c}{24} - h_2 - h_3} \\ & (16\mathfrak{q})^{h - \frac{c}{24}} \prod_{n\geq 1}(1-\mathfrak{q}^{2n})^{-1/2} H(h_1, h_2, h_3, h_4, h; \mathfrak{q}) \end{aligned} \tag{1.52}$$

with $H(h_1, h_2, h_3, h_4, h; \mathfrak{q})$ computed from the recursion relation (1.50). In this formalism, the recursion relation produce a series expansion of sphere 4 point block in $\mathfrak{q}$, which efficiently resum the series in z by

$$\mathfrak{q} = \frac{z}{16} + \frac{z^2}{32} \dots . \tag{1.53}$$

In practical application, h recursion product an efficient algorithm for computing the sphere 4 point block, which converges faster than the c recusion relation.

h recursion for sphere n point block in pillow frame

One can generalize the construction above to the sphere n point block. Put

$$\mathcal{F}_c(h_1, h_2, h_3, h_4, h; z) = \theta_3(0|\tau)^{\frac{c}{2}-4\sum_{i=1}^4 h_i} z^{\frac{c}{24}-h_1-h_2}(1-z)^{\frac{c}{24}-h_2-h_3} \prod_i \left(\frac{\partial w}{\partial x}(z_i)\right)^{-h_i} \left\langle \psi_{43} \middle| \mathbb{P}_h\, \mathfrak{q}^{L_0-\frac{c}{24}} \middle| \psi_{12} \right\rangle . \tag{1.54}$$

with 4 of them chosen to be at the corner of the pillow.

2 $\mathcal{N} = 1$ superconformal algebra

Let $\nu \in \{0, \frac{1}{2}\}$. The two-dimensional $\mathcal{N} = 1$ super-Virasoro algebra SCA_ν is generated by

$$\{L_n, G_r \mid n \in \mathbb{Z},\ r \in \mathbb{Z} + \nu\}.$$

Its nonvanishing (anti)commutation relations are

$$\begin{aligned} [L_m, L_n] &= (m-n)L_{m+n} + \frac{c}{12}\left(m^3 - m\right)\delta_{m+n,0}, \\ [L_m, G_r] &= \left(\frac{m}{2} - r\right) G_{m+r}, \\ \{G_r, G_s\} &= 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}. \end{aligned} \tag{2.1}$$

Here, $\nu = 0$ and $\nu = \frac{1}{2}$ correspond to the Ramond and Neveu–Schwarz sectors, respectively.

A two-dimensional $\mathcal{N} = (1, 1)$ superconformal field theory contains both holomorphic and anti-holomorphic superconformal algebras. We denote the anti-holomorphic algebra by $\widetilde{\mathsf{SCA}}_{\tilde{\nu}}$, with generators

$$\{\widetilde{L}_n, \widetilde{G}_r \mid n \in \mathbb{Z},\ r \in \mathbb{Z} + \tilde{\nu}\}.$$

They obey

$$\begin{aligned}
\left[\widetilde{L}_m, \widetilde{L}_n\right] &= (m-n)\widetilde{L}_{m+n} + \frac{\widetilde{c}}{12}\left(m^3 - m\right)\delta_{m+n,0},\\
\left[\widetilde{L}_m, \widetilde{G}_r\right] &= \left(\frac{m}{2} - r\right)\widetilde{G}_{m+r},\\
\left\{\widetilde{G}_r, \widetilde{G}_s\right\} &= 2\widetilde{L}_{r+s} + \frac{\widetilde{c}}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}.
\end{aligned} \tag{2.2}$$

The holomorphic and anti-holomorphic generators mutually (anti)commute.

3 Superconformal Block Decomposition

We now generalize this c-recursion relation to superconformal blocks in $\mathcal{N}=1$ superconformal field theories. The sphere 4-point case of the recursion relation has been worked out in [**0611266**, **0611295**], and the generalization to general plumbing channel has been worked out in [**1806.09563**].

Apart from the bosonic moduli, one also needs to specify a spin structure to put a SCFT on a Riemann surface. A plumbing channel is specified by a choice of NS and R for each punctures, a modulus q and a sign η for each pair of plumbed punctures. Only NS and NS, R and R punctures are allowed to be plumbed together, and the NS/R configuration of punctured on each 3-punctured sphere must be (NS, NS, NS) or (NS, R, R). For genus-g surface, there are 2^g choices of NS/R punctures and 2^g independent sign choices for each tube.

Similar to the notation $L_{-A}\nu$, we will denote a NS superconformal primary by ϕ, and its descendent by

$$\mathbf{L}_{-A}\phi \equiv L_{-n_1}\cdots L_{-n_\ell}G_{-r_1}\cdots G_{-r_p}\phi, \quad |A| \equiv \sum_{i=1}^{\ell} n_i + \sum_{j=1}^{p} r_j, \quad (-1)^A \equiv (-1)^p \tag{3.1}$$

where $n_1 \geq ... \geq n_\ell, r_1 > ... > r_p$. ϕ could be either Grassmann even or Grassmann odd.

In the language of 2D $\mathcal{N}=1$ superspace, one can pack ϕ and its level-$\frac{1}{2}$ descendent $\chi \equiv G_{-1/2}\phi$ in the primary superfield $\Phi(z,\theta)$, defined by

$$\Phi(z,\theta) = \phi(z) + \theta\chi(z). \tag{3.2}$$

Superconformal symmetry then constraints the 3-point functions to be of the following form

$$\begin{aligned}
\langle \Phi_1(z_1,\theta_1)\Phi_2(z_2,\theta_2)\rangle &\propto \frac{\delta_{h_1h_2}}{Z_{12}^{2h_1}},\\
\langle \Phi_1(z_1,\theta_1)\Phi_2(z_2,\theta_2)\Phi_3(z_3,\theta_3)\rangle &= \frac{\langle \Phi_1(0,0)\Phi_2(1,\Theta)\Phi_3'(\infty,0)\rangle}{Z_{21}^{h_1+h_2-h_3}Z_{31}^{h_1+h_3-h_2}Z_{32}^{h_2+h_3-h_1}},
\end{aligned} \tag{3.3}$$

where

$$
\begin{aligned}
Z_{ij} &\equiv z_i - z_j - \theta_i\theta_j \\
\Theta &\equiv \frac{\theta_1 z_{23} + \theta_2 z_{31} + \theta_3 z_{12} - \frac{1}{2}\theta_1\theta_2\theta_3}{\sqrt{z_{12}z_{13}z_{23}}}.
\end{aligned}
\tag{3.4}
$$

Importantly, there are now two independent structure constants for each triple of NS super-multiplets, they are the 1 and Θ component of the 3-point function, given essentially by $C_{123} = \langle\phi_1(0)\phi_2(1)\phi_3(\infty)\rangle$ and $\tilde{C}_{123} = \langle\phi_1(0)\chi_2(1)\phi_3(\infty)\rangle$. This is due to the fact that the 3-NS punctured genus-0 super-Riemann surface has 1 odd moduli, which is given precisely by this odd cross-ratio Θ.

For 3-point functions of superconformal descendants $\langle \mathbf{L}_{-A}\phi_1\mathbf{L}_{-B}\phi_2\mathbf{L}_{-C}\phi_3\rangle$, It can be shown by superconformal Ward identity that the $|A|+|B|+|C| \in \mathbb{Z}$ case only depends on C_{123}, while the $|A|+|B|+|C| \in \mathbb{Z}+\frac{1}{2}$ case only depends on $\tilde{C}_{123}$. We can then define two types of normalized 3-point functions

$$
\langle\langle \xi_3 \otimes \bar{\xi}_3 | \xi_2 \otimes \bar{\xi}_2(z,\bar{z}) | \xi_1 \otimes \bar{\xi}_1\rangle = (\pm)\begin{cases} C_{321}\rho_0(\xi_3,\xi_2,\xi_1|z)\rho_0(\bar{\xi}_3,\bar{\xi}_2,\bar{\xi}_1|\bar{z}), & \text{integer total level,} \\ \tilde{C}_{321}\rho_1(\xi_3,\xi_2,\xi_1|z)\rho_1(\bar{\xi}_3,\bar{\xi}_2,\bar{\xi}_1|\bar{z}), & \text{half-integer total level,}\end{cases}
\tag{3.5}
$$

by essentially normalizing

$$
\rho_0(\phi_1,\phi_2,\phi_3|1) = 1, \quad \rho_1(\phi_1,\chi_2,\phi_3|1) = 1.
\tag{3.6}
$$

The sign $(\pm)$ is here indicating that we use the operator order $\langle \xi_3\xi_2\xi_1\bar{\xi}_3\bar{\xi}_2\bar{\xi}_1\rangle$ to define the ρ on the RHS, which can differ from the operator order in LHS by a sign. Again, we denote $\rho_a(\xi_1,\xi_2,\xi_3) = \rho_a(\xi_1,\xi_2,\xi_3|1)$.

The anti-holomorphic sector is entirely analogous to the holomorphic sector. We distinguish quantities in the anti-holomorphic sector by adding a tilde over them. The standard BPZ bilinear pairing in a superconformal field theory would factorize into holomorphic and antiholomorphic pieces as

$$
\Big\langle\Big\langle \mathbf{L}_{-A}\phi \otimes \widetilde{\mathbf{L}}_{-\widetilde{A}}\widetilde{\phi} \;\Big|\; \mathbf{L}_{-B}\phi \otimes \widetilde{\mathbf{L}}_{-\widetilde{B}}\widetilde{\phi}\Big\rangle = (-1)^{(B+p_\phi)(\widetilde{A}+p_{\widetilde{\phi}})}\Big\langle\Big\langle \mathbf{L}_{-A}\phi \;\Big|\; \mathbf{L}_{-B}\phi\Big\rangle \times \Big\langle\Big\langle \widetilde{\mathbf{L}}_{-\widetilde{A}}\widetilde{\phi} \;\Big|\; \widetilde{\mathbf{L}}_{-\widetilde{B}}\widetilde{\phi}\Big\rangle.
\tag{3.7}
$$

3.1 Conformal Block Decomposition in All-NS Plumbing

For illustration, we consider a 2d $\mathcal{N} = (1,1)$ superconformal field theory and derive the conformal-block decomposition of its genus-2 partition function. The final result present a nice holomorphic factorialization structure after keeping track of the signs in the computation.

Let $\mathcal{O}_i(z,\bar{z})$ be the superconformal primary operator with conformal weight $(h_i, \tilde{h}_i)$. In the following derivation, for convenience, we will assume $\mathcal{O}_i$ to be orthogonal to each other, with

$$
\Big\langle\Big\langle O_i \;\Big|\; O_j\Big\rangle = D_i\delta_{ij}.
\tag{3.8}
$$

Here $\langle\!\langle\psi\big|$ denotes the BPZ conjugate of state ψ.

Primary operator $\mathcal{O}_i(z,\bar{z})$ admits a formal factorialization

$$\mathcal{O}_i(z,\bar{z}) = \sqrt{(-1)^{p_i\tilde{p}_i}D_i}\,\phi_i(z)\otimes\widetilde{\phi}_i(\bar{z}), \tag{3.9}$$

with ϕ the holomorphic superconformal primary with parity p_i and $\widetilde{\phi}(\bar{z})$ the anti-holomorphic superconformal primary with parity $\tilde{p}_i$, normalized such that

$$\langle\!\langle\phi \mid \phi\rangle = \langle\!\langle\widetilde{\phi} \mid \widetilde{\phi}\rangle = 1. \tag{3.10}$$

The parity of $\mathcal{O}_i$ is $p_i+\tilde{p}_i$. In the following, we will derive the conformal block decomposition for genus-two partition function $Z_{g=2}$ in theta channel and glass channel.

All-NS Block Decomposition in Theta Channel

Consider the genus-two partition function $Z_{g=2}$. We work in the theta-graph plumbing channel, taking all three tubes to have (NS,NS) spin structure in both the holomorphic and anti-holomorphic sectors. In the theta plumbing channel, the genus-two partition function $Z_{g=2}$ has a prescription

$$\begin{aligned}
Z^{\Theta}_{g=2} = &\sum_{\mathcal{O}_1,\mathcal{O}_2,\mathcal{O}_3}\ \sum_{\substack{|A|=|B|,\ |C|=|D|,\ |E|=|F|\\ |\widetilde{A}|=|\widetilde{B}|,\ |\widetilde{C}|=|\widetilde{D}|,\ |\widetilde{E}|=|\widetilde{F}|}}\ \prod_{i=1}^{3} q_i^{h_i}\widetilde{q}_i^{\widetilde{h}_i}\eta_i^{p_i}\widetilde{\eta}_i^{\tilde{p}_i} \times q_1^{|A|}\eta_1^{A}q_2^{|C|}\eta_2^{C}q_3^{|E|}\eta_3^{E}\widetilde{q}_1^{|\widetilde{A}|}\widetilde{\eta}_1^{\widetilde{A}}\widetilde{q}_2^{|\widetilde{C}|}\widetilde{\eta}_2^{\widetilde{C}}\widetilde{q}_3^{|\widetilde{E}|}\widetilde{\eta}_3^{\widetilde{E}}\\
&\times(-1)^{(A+\widetilde{A}+p_1+\tilde{p}_1)(C+\widetilde{C}+p_2+\tilde{p}_2)+(A+\widetilde{A}+p_1+\tilde{p}_1)(E+\widetilde{E}+p_3+\tilde{p}_3)+(C+\widetilde{C}+p_2+\tilde{p}_2)(E+\widetilde{E}+p_3+\tilde{p}_3)}\\
&\times(\mathbb{G}_1)^{A\widetilde{A};B\widetilde{B}}(\mathbb{G}_2)^{C\widetilde{C};D\widetilde{D}}(\mathbb{G}_3)^{E\widetilde{E};F\widetilde{F}}\\
&\times\left\langle\!\!\left\langle\mathbf{L}_{-A}\widetilde{\mathbf{L}}_{-\widetilde{A}}\mathcal{O}_1\middle|\mathbf{L}_{-C}\widetilde{\mathbf{L}}_{-\widetilde{C}}\mathcal{O}_2(1,1)\middle|\mathbf{L}_{-E}\widetilde{\mathbf{L}}_{-\widetilde{E}}\mathcal{O}_3\right\rangle\right. \times \left\langle\!\!\left\langle\mathbf{L}_{-B}\widetilde{\mathbf{L}}_{-\widetilde{B}}\mathcal{O}_1\middle|\mathbf{L}_{-D}\widetilde{\mathbf{L}}_{-\widetilde{D}}\mathcal{O}_2(1,1)\middle|\mathbf{L}_{-F}\widetilde{\mathbf{L}}_{-\widetilde{F}}\mathcal{O}_3\right\rangle\right.
\end{aligned} \tag{3.11}$$

In this prescription, we need to choose the Weyl frame to be the one defined by theta channel plumbing. The sign in the sum comes from reordering the operators into the prescription of the matrix element. The gram matrix is defined by

$$(\mathbb{G}_i)_{A\widetilde{A};B\widetilde{B}} \equiv \langle\!\langle\mathbf{L}_{-A}\widetilde{\mathbf{L}}_{-\widetilde{A}}\mathcal{O}_i \mid \mathbf{L}_{-B}\widetilde{\mathbf{L}}_{-\widetilde{B}}\mathcal{O}_i\rangle \tag{3.12}$$

It can be factorized into the holomorphic and anti-holomorphic pieces as

$$(\mathbb{G}_i)_{A\widetilde{A};B\widetilde{B}} = D_i(-1)^{(\widetilde{A}+\tilde{p}_i)(B+p_i)+p_i\tilde{p}_i}(\mathbf{G}_{h_i})_{AB}(\widetilde{\mathbf{G}}_{\widetilde{h}_i})_{\widetilde{A}\widetilde{B}}. \tag{3.13}$$

where

$$(\mathbf{G}_{h_i})_{AB} := \langle\!\langle\mathbf{L}_{-A}\phi_i \mid \mathbf{L}_{-B}\phi_i\rangle,\ (\widetilde{\mathbf{G}}_{h_i})_{\widetilde{A}\widetilde{B}} := \langle\!\langle\widetilde{\mathbf{L}}_{-\widetilde{A}}\tilde{\phi}_i \mid \widetilde{\mathbf{L}}_{-\widetilde{B}}\tilde{\phi}_i\rangle. \tag{3.14}$$

The three point function would also factorize

$$
\begin{aligned}
&\left\langle\!\!\left\langle \mathbf{L}_{-A}\widetilde{\mathbf{L}}_{-\widetilde{A}}\mathcal{O}_1 \middle| \mathbf{L}_{-C}\widetilde{\mathbf{L}}_{-\widetilde{C}}\mathcal{O}_2(1,1) \middle| \mathbf{L}_{-E}\widetilde{\mathbf{L}}_{-\widetilde{E}}\mathcal{O}_3 \right\rangle\right. \\
&\quad \times \left\langle\!\!\left\langle \mathbf{L}_{-B}\widetilde{\mathbf{L}}_{-\widetilde{B}}\mathcal{O}_1 \middle| \mathbf{L}_{-D}\widetilde{\mathbf{L}}_{-\widetilde{D}}\mathcal{O}_2(1,1) \middle| \mathbf{L}_{-F}\widetilde{\mathbf{L}}_{-\widetilde{F}}\mathcal{O}_3 \right\rangle\right. \\
&= (-1)^{p_1\tilde{p}_1+p_2\tilde{p}_2+p_3\tilde{p}_3} D_1 D_2 D_3 \left(C_{123}^{(a)}\right)^2 \\
&\quad \times \rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3)\, \widetilde{\rho}_{\tilde{a}}\left(\widetilde{\mathbf{L}}_{-\widetilde{A}}\widetilde{\phi}_1, \widetilde{\mathbf{L}}_{-\widetilde{C}}\widetilde{\phi}_2, \widetilde{\mathbf{L}}_{-\widetilde{E}}\widetilde{\phi}_3\right) \\
&\quad \times \rho_a(\mathbf{L}_{-B}\phi_1, \mathbf{L}_{-D}\phi_2, \mathbf{L}_{-F}\phi_3)\, \widetilde{\rho}_{\tilde{a}}\left(\widetilde{\mathbf{L}}_{-\widetilde{B}}\widetilde{\phi}_1, \widetilde{\mathbf{L}}_{-\widetilde{D}}\widetilde{\phi}_2, \widetilde{\mathbf{L}}_{-\widetilde{F}}\widetilde{\phi}_3\right).
\end{aligned} \tag{3.15}
$$

Here $a = A+C+E \pmod 2, \tilde{a} = \tilde{A}+\tilde{C}+\tilde{E} \pmod 2$. It satisfy $a+p_1+p_2+p_3 = \tilde{a}+\tilde{p_1}+\tilde{p_2}+\tilde{p_3} \pmod 2$. In this step, the sign comming from the reordering of holomorphic primaries and anti-holomorphic primaries will cancel out because the same factor appears in both matrix element.

Using the factorialization properies, the genus 2 partition function simplifies into

$$
\begin{aligned}
Z_{g=2} = &\sum_{\mathcal{O}_1,\mathcal{O}_2,\mathcal{O}_3} \sum_{\substack{|A|=|B|,\ |C|=|D|,\ |E|=|F| \\ |\widetilde{A}|=|\widetilde{B}|,\ |\widetilde{C}|=|\widetilde{D}|,\ |\widetilde{E}|=|\widetilde{F}|}} \prod_{i=1}^{3} q_i^{h_i}\widetilde{q}_i^{\widetilde{h}_i}\eta_i^{p_i}\widetilde{\eta}_i^{\widetilde{p}_i} \times q_1^{|A|}\eta_1^{A} q_2^{|C|}\eta_2^{C} q_3^{|E|}\eta_3^{E} \widetilde{q}_1^{|\widetilde{A}|}\widetilde{\eta}_1^{\widetilde{A}} \widetilde{q}_2^{|\widetilde{C}|}\widetilde{\eta}_2^{\widetilde{C}} \widetilde{q}_3^{|\widetilde{E}|}\widetilde{\eta}_3^{\widetilde{E}} \\
&\times (-1)^{a+p_1+p_2+p_3+K+\bar{K}} \left(C_{123}^{(a)}\right)^2 \mathbf{G}_{h_1}^{AB}\mathbf{G}_{h_2}^{CD}\mathbf{G}_{h_3}^{EF}\widetilde{\mathbf{G}}_{\widetilde{h}_1}^{\widetilde{A}\widetilde{B}}\widetilde{\mathbf{G}}_{\widetilde{h}_2}^{\widetilde{C}\widetilde{D}}\widetilde{\mathbf{G}}_{\widetilde{h}_3}^{\widetilde{E}\widetilde{F}} \\
&\times \rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3)\, \widetilde{\rho}_{\tilde{a}}\left(\widetilde{\mathbf{L}}_{-\widetilde{A}}\widetilde{\phi}_1, \widetilde{\mathbf{L}}_{-\widetilde{C}}\widetilde{\phi}_2, \widetilde{\mathbf{L}}_{-\widetilde{E}}\widetilde{\phi}_3\right) \\
&\times \rho_a(\mathbf{L}_{-B}\phi_1, \mathbf{L}_{-D}\phi_2, \mathbf{L}_{-F}\phi_3)\, \widetilde{\rho}_{\tilde{a}}\left(\widetilde{\mathbf{L}}_{-\widetilde{B}}\widetilde{\phi}_1, \widetilde{\mathbf{L}}_{-\widetilde{D}}\widetilde{\phi}_2, \widetilde{\mathbf{L}}_{-\widetilde{F}}\widetilde{\phi}_3\right).
\end{aligned} \tag{3.16}
$$

Here

$$
\begin{aligned}
K &\equiv (A+p_1)(C+p_2) + (A+p_1)(E+p_3) + (C+p_2)(E+p_3) \pmod 2, \\
\bar{K} &\equiv (\widetilde{A}+\widetilde{p}_1)(\widetilde{C}+\widetilde{p}_2) + (\widetilde{A}+\widetilde{p}_1)(\widetilde{E}+\widetilde{p}_3) + (\widetilde{C}+\widetilde{p}_2)(\widetilde{E}+\widetilde{p}_3) \pmod 2.
\end{aligned} \tag{3.17}
$$

Define the NS superconformal block by

$$
\begin{aligned}
\mathbf{F}_a^{\Theta}(h_1,h_2,h_3,c) = &\sum_{\substack{|A|=|B|,|C|=|D|,|E|=|F|, \\ (-1)^{A+C+E}=(-1)^a}} q_1^{|A|}\eta_1^{A+p_1} q_2^{|C|}\eta_2^{C+p_2} q_3^{|E|}\eta_3^{E+p_3} \mathbf{G}_{h_1}^{AB}\mathbf{G}_{h_2}^{CD}\mathbf{G}_{h_3}^{EF} \\
&\times (-1)^K \rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3)\rho_a(\mathbf{L}_{-B}\phi_1, \mathbf{L}_{-D}\phi_2, \mathbf{L}_{-F}\phi_3),
\end{aligned} \tag{3.18}
$$

and analogously the anti-holomorphic block $\widetilde{F}_{\tilde{a}}$. The genus 2 partition function then reduces to

$$
Z_{g=2} = \sum_{\mathcal{O}_1,\mathcal{O}_2,\mathcal{O}_3} \sum_{a=0,1} \left(C_{123}^{(a)}\right)^2 (-1)^{a+p_1+p_2+p_3} \prod_{i=1}^{3} q_i^{h_i}\widetilde{q}_i^{\widetilde{h}_i} F_a(h_1,h_2,h_3,c)\widetilde{F}_{\tilde{a}}(\tilde{h}_1,\tilde{h}_2,\tilde{h}_3,\tilde{c}). \tag{3.19}
$$

where $a+p_1+p_2+p_3 = \tilde{a}+\tilde{p_1}+\tilde{p_2}+\tilde{p_3} \pmod 2$.

All-NS Block Decomposition in Glass Channel

We can also compute the genus 2 partition function in the glass channel. It is given by the following prescription

$$
\begin{aligned}
Z_{g=2}^{Gl} = &\sum_{\mathcal{O}_1,\mathcal{O}_2,\mathcal{O}_3} \sum_{\substack{|A|=|B|,\ |C|=|D|,\ |E|=|F| \\ |\widetilde{A}|=|\widetilde{B}|,\ |\widetilde{C}|=|\widetilde{D}|,\ |\widetilde{E}|=|\widetilde{F}|}} \prod_{i=1}^{3} q_i^{h_i} \widetilde{q}_i^{\widetilde{h}_i} \eta_i^{p_i} \widetilde{\eta}_i^{\widetilde{p}_i} \times q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E \widetilde{q}_1^{|\widetilde{A}|} \widetilde{\eta}_1^{\widetilde{A}} \widetilde{q}_2^{|\widetilde{C}|} \widetilde{\eta}_2^{\widetilde{C}} \widetilde{q}_3^{|\widetilde{E}|} \widetilde{\eta}_3^{\widetilde{E}} \\
&\times (-1)^{(E+\widetilde{E}+p_3+\widetilde{p}_3)\left[(A+\widetilde{A}+p_1+\widetilde{p}_1)+(C+\widetilde{C}+p_2+\widetilde{p}_2)\right]} \\
&\times (\mathbb{G}_1)^{A\widetilde{A};B\widetilde{B}} (\mathbb{G}_2)^{C\widetilde{C};D\widetilde{D}} (\mathbb{G}_3)^{E\widetilde{E};F\widetilde{F}} \\
&\times \left\langle\!\left\langle \mathbf{L}_{-A}\widetilde{\mathbf{L}}_{-\widetilde{A}}\mathcal{O}_1 \middle| \mathbf{L}_{-E}\widetilde{\mathbf{L}}_{-\widetilde{E}}\mathcal{O}_3(1,1) \middle| \mathbf{L}_{-B}\widetilde{\mathbf{L}}_{-\widetilde{B}}\mathcal{O}_1 \right\rangle\right. \times \left\langle\!\left\langle \mathbf{L}_{-C}\widetilde{\mathbf{L}}_{-\widetilde{C}}\mathcal{O}_2 \middle| \mathbf{L}_{-F}\widetilde{\mathbf{L}}_{-\widetilde{F}}\mathcal{O}_3(1,1) \middle| \mathbf{L}_{-D}\widetilde{\mathbf{L}}_{-\widetilde{D}}\mathcal{O}_2 \right\rangle\right.
\end{aligned} \tag{3.20}
$$

The gram matrix factorize in the same way as (3.13). The three point function would factorize as

$$
\begin{aligned}
&\left\langle\!\left\langle \mathbf{L}_{-A}\widetilde{\mathbf{L}}_{-\widetilde{A}}\mathcal{O}_1 \middle| \mathbf{L}_{-E}\widetilde{\mathbf{L}}_{-\widetilde{E}}\mathcal{O}_3(1,1) \middle| \mathbf{L}_{-B}\widetilde{\mathbf{L}}_{-\widetilde{B}}\mathcal{O}_1 \right\rangle\right. \\
&\quad \times \left\langle\!\left\langle \mathbf{L}_{-C}\widetilde{\mathbf{L}}_{-\widetilde{C}}\mathcal{O}_2 \middle| \mathbf{L}_{-F}\widetilde{\mathbf{L}}_{-\widetilde{F}}\mathcal{O}_3(1,1) \middle| \mathbf{L}_{-D}\widetilde{\mathbf{L}}_{-\widetilde{D}}\mathcal{O}_2 \right\rangle\right. \\
&= (-1)^{p_1\widetilde{p}_1+p_2\widetilde{p}_2+p_3\widetilde{p}_3+(A+p_1+E+p_3)(\widetilde{A}+\widetilde{p}_1+\widetilde{E}+\widetilde{p}_3)+(C+p_2+E+p_3)(\widetilde{C}+\widetilde{p}_2+\widetilde{E}+\widetilde{p}_3)} D_1 D_2 D_3 \left(C_{131}^{(a)} C_{232}^{(a)}\right) \\
&\quad \times \rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-E}\phi_3, \mathbf{L}_{-B}\phi_1)\, \widetilde{\rho}_{\tilde{a}}\left(\widetilde{\mathbf{L}}_{-\widetilde{A}}\widetilde{\phi}_1, \widetilde{\mathbf{L}}_{-\widetilde{E}}\widetilde{\phi}_3, \widetilde{\mathbf{L}}_{-\widetilde{B}}\widetilde{\phi}_1\right) \\
&\quad \times \rho_a(\mathbf{L}_{-C}\phi_2, \mathbf{L}_{-F}\phi_3, \mathbf{L}_{-D}\phi_2)\, \widetilde{\rho}_{\tilde{a}}\left(\widetilde{\mathbf{L}}_{-\widetilde{C}}\widetilde{\phi}_2, \widetilde{\mathbf{L}}_{-\widetilde{F}}\widetilde{\phi}_3, \widetilde{\mathbf{L}}_{-\widetilde{D}}\widetilde{\phi}_2\right).
\end{aligned} \tag{3.21}
$$

where $a \equiv E \pmod 2$ and $\tilde{a} = \widetilde{E} \pmod 2$.

Using the factorialization properties, the glass channel genus 2 partition function can be expressed into

$$
\begin{aligned}
Z_{g=2}^{\mathrm{Gl}} = &\sum_{\mathcal{O}_1,\mathcal{O}_2,\mathcal{O}_3} \sum_{\substack{|A|=|B|,\ |C|=|D|,\ |E|=|F| \\ |\widetilde{A}|=|\widetilde{B}|,\ |\widetilde{C}|=|\widetilde{D}|,\ |\widetilde{E}|=|\widetilde{F}|}} \prod_{i=1}^{3} q_i^{h_i} \widetilde{q}_i^{\widetilde{h}_i} \eta_i^{p_i} \widetilde{\eta}_i^{\widetilde{p}_i} q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E \widetilde{q}_1^{|\widetilde{A}|} \widetilde{\eta}_1^{\widetilde{A}} \widetilde{q}_2^{|\widetilde{C}|} \widetilde{\eta}_2^{\widetilde{C}} \widetilde{q}_3^{|\widetilde{E}|} \widetilde{\eta}_3^{\widetilde{E}} \\
&\times C_{131}^{(a)} C_{232}^{(a)}\, (-1)^{a+p_3+K_{\mathrm{Gl}}+\widetilde{K}_{\mathrm{Gl}}}\, \mathbf{G}_{h_1}^{AB} \mathbf{G}_{h_2}^{CD} \mathbf{G}_{h_3}^{EF} \widetilde{\mathbf{G}}_{\widetilde{h}_1}^{\widetilde{A}\widetilde{B}} \widetilde{\mathbf{G}}_{\widetilde{h}_2}^{\widetilde{C}\widetilde{D}} \widetilde{\mathbf{G}}_{\widetilde{h}_3}^{\widetilde{E}\widetilde{F}} \\
&\times \rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-E}\phi_3, \mathbf{L}_{-B}\phi_1)\, \rho_a(\mathbf{L}_{-C}\phi_2, \mathbf{L}_{-F}\phi_3, \mathbf{L}_{-D}\phi_2) \\
&\times \widetilde{\rho}_a\left(\widetilde{\mathbf{L}}_{-\widetilde{A}}\widetilde{\phi}_1, \widetilde{\mathbf{L}}_{-\widetilde{E}}\widetilde{\phi}_3, \widetilde{\mathbf{L}}_{-\widetilde{B}}\widetilde{\phi}_1\right) \widetilde{\rho}_a\left(\widetilde{\mathbf{L}}_{-\widetilde{C}}\widetilde{\phi}_2, \widetilde{\mathbf{L}}_{-\widetilde{F}}\widetilde{\phi}_3, \widetilde{\mathbf{L}}_{-\widetilde{D}}\widetilde{\phi}_2\right).
\end{aligned}
$$

where

$$
K_{\mathrm{Gl}} = a\big(A + p_1 + C + p_2\big), \quad \widetilde{K}_{\mathrm{Gl}} = a\big(\widetilde{A} + \widetilde{p}_1 + \widetilde{C} + \widetilde{p}_2\big). \tag{3.22}
$$

Define the NS superconformal block in glass channel as

$$\begin{aligned}\mathbf{F}_a^{\mathrm{Gl}}(h_1,h_2,h_3,c) = &\sum_{|A|=|B|,|C|=|D|,|E|=|F|,(-1)^E=(-1)^a} q_1^{|A|}\eta_1^{A+p_1}q_2^{|C|}\eta_2^{C+p_2}q_3^{|E|}\eta_3^{E+p_3}\mathbf{G}_{h_1}^{AB}\mathbf{G}_{h_2}^{CD}\mathbf{G}_{h_3}^{EF}\\ &\times(-1)^{K_{\mathrm{Gl}}}\rho_a(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-E}\phi_3,\mathbf{L}_{-B}\phi_1)\rho_a(\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-F}\phi_3,\mathbf{L}_{-D}\phi_2).\end{aligned} \tag{3.23}$$

and analogously the anti-holomorphic block $\widetilde{F}_a^{\mathrm{Gl}}$. The genus 2 partition function then reduces to

$$Z_{g=2} = \sum_{\mathcal{O}_1,\mathcal{O}_2,\mathcal{O}_3}\sum_{a=0,1}\left(C_{131}^{(a)}C_{232}^{(a)}\right)(-1)^{a+p_1}\prod_{i=1}^{3} q_i^{h_i}\tilde{q}_i^{\tilde{h}_i}F_a^{\mathrm{Gl}}(h_1,h_2,h_3,c)\widetilde{F}_{\tilde{a}}^{\mathrm{Gl}}(\tilde{h}_1,\tilde{h}_2,\tilde{h}_3,\tilde{c}). \tag{3.24}$$

where $a+p_1=\tilde{a}+\tilde{p}_1 \pmod 2$.

3.2 Conformal Block Decomposition in mixed NS/R Plumbing

4 NS superconformal blocks

4.1 c-Recursion for NS Block

We will keep using the theta channel as an example, with the plumbing data denoted by (q_i,η_i). The all-NS conformal block is given by

$$\begin{aligned}\mathbf{F}_a(h_1,h_2,h_3,c) = &\sum_{\substack{|A|=|B|,\ |C|=|D|,\ |E|=|F|\\ (-1)^{A+C+E}=(-1)^a}} q_1^{|A|}\eta_1^{A+p_1}q_2^{|C|}\eta_2^{C+p_2}q_3^{|E|}\eta_3^{E+p_3}\mathbf{G}_{h_1}^{AB}\mathbf{G}_{h_2}^{CD}\mathbf{G}_{h_3}^{EF}\\ &\times(-1)^{(A+p_1)(C+p_2)+(A+p_1)(E+p_3)+(C+p_2)(E+p_3)}\\ &\rho_a(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-E}\phi_3)\rho_a(\mathbf{L}_{-B}\phi_1,\mathbf{L}_{-D}\phi_2,\mathbf{L}_{-F}\phi_3),\end{aligned} \tag{4.1}$$

where $\mathbf{G}$ denotes the inverse of the super-Gram matrix $\langle h|\mathbf{L}_{-A}^\dagger\mathbf{L}_{-B}|h\rangle$. a takes value from $0,1$. The sign $(-1)^{(A+p_1)(C+p_2)+(A+p_1)(E+p_3)+(C+p_2)(E+p_3)}$ comes from moving the operator orders. The condition $(-1)^{A+C+E}=(-1)^a$ ensures that only total-integer level or total-half-integer level intermediate states enters the 3-point function.

The NS supermultiplet admits a level-$rs/2$ null state when

$$h_{rs}(c) = \frac{Q^2-(rb+sb^{-1})^2}{8}, \quad Q=b+b^{-1} \quad (r\geq 2, s\geq 1,\ r+s\in 2\mathbb{Z}), \tag{4.2}$$

where b is defined by $c=\frac{3}{2}+3(b+b^{-1})^2$. Conversely, for fixed h, the degeneration happens when the central charge is

$$c_{r,s}(h) = \frac{15}{2} + 3(x_{rs} + x_{rs}^{-1}), \quad b_{rs} = \sqrt{x_{rs}}, \quad x_{rs} = \frac{rs - 1 + 4h + \sqrt{(r-s)^2 + 8(rs-1)h + 16h^2}}{1 - r^2}. \tag{4.3}$$

At $c = c_{r,s}$, the Gram matrix becomes degenerate, which indicate a pole in the conformal block. For generic h_i, the pole is a simple pole. It suggests the following structure for the all-NS conformal block

$$\mathbf{F}_a(h_1, h_2, h_3, c) = \mathbf{F}_a^{\text{global}}(h_1, h_2, h_3, c) + \sum_{i=1}^{3} \sum_{r \geq 2, s \geq 1} \frac{\mathbf{W}_{a,rs}^{(i)}(h_1, h_2, h_3)}{c - c_{r,s}(h_i)} \tag{4.4}$$

where $\mathbf{F}_a^{\text{global}}(h_1, h_2, h_3, c)$ is regular on the entire c plane. Follow from the representation analysis as for the Virasoro block, we can determine that $\mathbf{F}_a^{\text{global}}(h_1, h_2, h_3, c)$ is also regular at $c \to \infty$. It determines that $\mathbf{F}_a^{\text{global}}(h_1, h_2, h_3, c)$ must be independent of c. Therefore, we can access the $\mathbf{F}_a^{\text{global}}$ by taking the limit

$$\mathbf{F}_a^{\text{global}}(h_1, h_2, h_3) = \lim_{c \to \infty} \mathbf{F}_a(h_1, h_2, h_3, c). \tag{4.5}$$

It would be referred to as the large-c seed term.

The subalgebra of $\mathcal{N} = 1$ SCA that's independent of c is given by the $\mathfrak{osp}(1|2)$ algebra generated by $\{L_{0,\pm 1}, G_{\pm 1/2}\}$. We thus define $\hat{A}$ as the number of non-$L_{-1}, G_{-1/2}$ elements in $\mathbf{L}_{-A}$. The same c-power counting argument works also in the SUSY case: If we construct a orthonormal basis $\mathcal{L}_{-A}$ by applying Gram-Schmidt on $\mathbf{L}_{-A}$, then in the large c limit the 3-point function $\rho(\mathcal{L}_{-A}\phi_1, \mathcal{L}_{-B}\phi_2, \mathcal{L}_{-C}\phi_3)$ factorizes into the large-c vacuum block and the $\mathfrak{osp}(1|2)$ block,

$$\lim_{c \to \infty} \mathbf{F}_a(h_1, h_2, h_3, c) = \sum_{\substack{a_1, a_2, a_3 \in \{0,1\} \\ a_1 + a_2 + a_3 \equiv a \pmod 2}} \mathbf{F}_0(0, 0, 0, \infty; \eta_i \to (-1)^{a_i + p_i} \eta_i) \times \mathbf{F}_{a_1 a_2 a_3}^{\mathfrak{osp}(1|2)}(h_1, h_2, h_3), \tag{4.6}$$

where the $\mathfrak{osp}(1|2)$ block is defined for $a_i \in \{0, 1\}$ and $a_1 + a_2 + a_3 \equiv a \pmod 2$ by

$$\begin{aligned} \mathbf{F}_{a_1 a_2 a_3}^{\mathfrak{osp}(1|2)}(h_1, h_2, h_3) = \sum_{k_1, k_2, k_3} & q_1^{k_1 + \frac{1}{2}a_1} \eta_1^{p_1 + a_1} q_2^{k_2 + \frac{1}{2}a_2} \eta_2^{p_2 + a_2} q_3^{k_3 + \frac{1}{2}a_3} \eta_3^{p_3 + a_3} \\ & \times (-1)^{(a_1+p_1)(a_2+p_2) + (a_1+p_1)(a_3+p_3) + (a_2+p_2)(a_3+p_3)} \\ & \times \left[\prod_{i=1}^{3} \frac{1}{\| L_{-1}^{k_i} G_{-\frac{1}{2}}^{a_i} \phi_i \|^2} \right] \rho_a(L_{-1}^{k_1} G_{-\frac{1}{2}}^{a_1} \phi_1, L_{-1}^{k_2} G_{-\frac{1}{2}}^{a_2} \phi_2, L_{-1}^{k_3} G_{-\frac{1}{2}}^{a_3} \phi_3)^2 \end{aligned} \tag{4.7}$$

Importantly, the large-c limit of the odd vacuum block is 0, due to the fact that one can only contract G_r, G_{-r} pairwise. Due to the Grassmann parity of $G_{-1/2}^a \phi$, for each fixed a one must sum over the corresponding 4 configurations of global $\mathfrak{sl}(2)$ primaries independently; the two values of a give 8 configurations in total. The sign η_i in the corresponding global block must be shifted. Similar to the Virasoro case, the vacuum block can also be directly computed in Schottky parameters. We will focus on the global $\mathfrak{osp}(1|2)$ blocks in the next section.

4.2 Global $\mathfrak{osp}(1|2)$ blocks

In this section, we develop the method of computing arbitrary Global $\mathfrak{osp}(1|2)$ blocks. We take the following basis for states

$$L_{-1}^k G_{-\frac{1}{2}}^a \phi, \quad k \in \mathbb{N},\ a \in \{0,1\}. \tag{4.8}$$

with ϕ normalized such that $\|\phi\|^2 = 1$. The nontrivial inner products are

$$\|L_{-1}^k \phi\|^2 = k!(2h)_k, \quad \|L_{-1}^k G_{-\frac{1}{2}}\phi\|^2 = k!(2h)_{k+1}, \tag{4.9}$$

where $(x)_n$ is the rising Pochhammer symbol.

For the genus-2 block in the theta channel, fix $a_i \in \{0,1\}$ and set $a \equiv a_1 + a_2 + a_3 \pmod 2$. We then have

$$\begin{aligned}\mathbf{F}^{\mathfrak{osp}(1|2)}_{a_1a_2a_3}(h_1,h_2,h_3) = &\sum_{k_1,k_2,k_3} q_1^{k_1+\frac{1}{2}a_1} \eta_1^{a_1+p_1} q_2^{k_2+\frac{1}{2}a_2} \eta_2^{a_2+p_2} q_3^{k_3+\frac{1}{2}a_3} \eta_3^{a_3+p_3} \\ &\times (-1)^{(a_1+p_1)(a_2+p_2)+(a_1+p_1)(a_3+p_3)+(a_2+p_2)(a_3+p_3)} \\ &\times \left[\prod_{i=1}^{3} \frac{1}{k_i!(2h_i)_{k_i+a_i}}\right] \rho_a(L_{-1}^{k_1}G_{-\frac{1}{2}}^{a_1}\phi_1, L_{-1}^{k_2}G_{-\frac{1}{2}}^{a_2}\phi_2, L_{-1}^{k_3}G_{-\frac{1}{2}}^{a_3}\phi_3)^2\end{aligned} \tag{4.10}$$

The remaining object to compute is simply $\rho_a(L_{-1}^{k_1}G_{-\frac{1}{2}}^{a_1}\phi_1, L_{-1}^{k_2}G_{-\frac{1}{2}}^{a_2}\phi_2, L_{-1}^{k_3}G_{-\frac{1}{2}}^{a_3}\phi_3)$, which can be computed directly since $G^a_{-\frac{1}{2}}\phi$ is a $\mathfrak{sl}(2)$ primary and one can use the result for $\mathfrak{sl}(2)$ 3-point functions. We keep the normalization

$$\rho_0(\phi_1,\phi_2,\phi_3) = 1, \quad \rho_1(\phi_1, G_{-1/2}\phi_2, \phi_3) = 1. \tag{4.11}$$

Superconformal Wald identity fix the rest to be

$(a_1a_2a_3)$	a	$\rho_a\left(G^{a_1}_{-\frac{1}{2}}\phi_1, G^{a_2}_{-\frac{1}{2}}\phi_2, G^{a_3}_{-\frac{1}{2}}\phi_3\right)$
000	0	1
110	0	$(-1)^{p_1}(h_1+h_2-h_3)$
101	0	$(-1)^{p_1+p_2}(h_1-h_2+h_3)$
011	0	$(-1)^{p_2}(h_1-h_2-h_3)$
100	1	$(-1)^{p_1}$
010	1	1
001	1	$(-1)^{p_2+1}$
111	1	$(-1)^{p_1+p_2+1}\left(h_1+h_2+h_3-\frac{1}{2}\right)$

Defining

$$s_{km}(h_1,h_2,h_3) = \sum_{p=0}^{\min(k,m)} \frac{k!}{p!(k-p)!}(2h_3+m-1)^{(p)} m^{(p)} \\ \times (h_3+h_2-h_1)_{m-p}(h_1+h_2-h_3+p-m)_{k-p}, \tag{4.12}$$

where $(x)^{(n)}$ is the falling Pochhammer symbol. Using the formula for $\mathfrak{sl}(2)$ 3-point functions in [**Cho-Collier-Yin**], one then arrive at

$$\begin{aligned}&\rho_a\Big(L_{-1}^{k_1}G_{-1/2}^{a_1}\phi_1, L_{-1}^{k_2}G_{-1/2}^{a_2}\phi_2, L_{-1}^{k_3}G_{-1/2}^{a_3}\phi_3\Big)\\ &\quad= \rho_a\Big(G_{-1/2}^{a_1}\phi_1, G_{-1/2}^{a_2}\phi_2, G_{-1/2}^{a_3}\phi_3\Big)\, s_{k_1k_3}\Big(h_1+\frac{a_1}{2}, h_2+\frac{a_2}{2}, h_3+\frac{a_3}{2}\Big)\\ &\qquad\times\Big(h_1+\frac{a_1}{2}+k_1-h_2-\frac{a_2}{2}-k_2+1-h_3-\frac{a_3}{2}-k_3\Big)_{k_2},\end{aligned} \tag{4.13}$$

and hence

$$\begin{aligned}\mathbf{F}^{\mathfrak{osp}(1|2)}_{a_1a_2a_3}(h_1,h_2,h_3) = &\sum_{k_1,k_2,k_3} q_1^{k_1+\frac{1}{2}a_1}\eta_1^{a_1+p_1} q_2^{k_2+\frac{1}{2}a_2}\eta_2^{a_2+p_2} q_3^{k_3+\frac{1}{2}a_3}\eta_3^{a_3+p_3}\\ &\times(-1)^{(a_1+p_1)(a_2+p_2)+(a_1+p_1)(a_3+p_3)+(a_2+p_2)(a_3+p_3)}\left[\prod_{i=1}^{3}\frac{1}{k_i!(2h_i)_{k_i+a_i}}\right]\\ &\times\Bigg[\rho_a\Big(G_{-1/2}^{a_1}\phi_1, G_{-1/2}^{a_2}\phi_2, G_{-1/2}^{a_3}\phi_3\Big)\, s_{k_1k_3}\Big(h_1+\frac{a_1}{2}, h_2+\frac{a_2}{2}, h_3+\frac{a_3}{2}\Big)\\ &\qquad\times\Big(h_1+\frac{a_1}{2}+k_1-h_2-\frac{a_2}{2}-k_2+1-h_3-\frac{a_3}{2}-k_3\Big)_{k_2}\Bigg]^2.\end{aligned} \tag{4.14}$$

Using this formula, one can compute the $\mathfrak{osp}(1|2)$ block directly as a q-expansion. This concludes our discussion for the all-NS c-recursion relation.

4.3 Residue in the NS Block c-Recursion

At a pole c_{rs} on the complex c-plane, the residues are contributed by the descendents of this null state, which we will call $\chi_{rs} = \sum_M \chi^M_{rs} \mathbf{L}_{-M}\phi$. The pole of $\mathbf{F}_a$ at $c = c_{rs}(h_1)$ is given by

$$
\begin{aligned}
\mathbf{W}^{(1)}_{a,rs} = &-\frac{\partial c_{rs}(h_1)}{\partial h_1} \lim_{h_1\to h_{r,s}} \frac{h_1 - h_{rs}}{\| \sum_M \chi^M_{rs}\mathbf{L}_{-M}\phi_1\|^2} \\
&\times \sum_{\substack{|A|=|B|,\ |C|=|D|,\ |E|=|F| \\ (-1)^{A+C+E}=(-1)^{a+rs}}} q_1^{|A|+\frac{rs}{2}} \eta_1^{A+p_1+rs} q_2^{|C|}\eta_2^{C+p_2} q_3^{|E|}\eta_3^{E+p_3} \mathbf{G}^{AB}_{h_1+\frac{rs}{2}} \mathbf{G}^{CD}_{h_2} \mathbf{G}^{EF}_{h_3} \\
&\qquad (-1)^{(A+p_1+rs)(C+p_2)+(A+p_1+rs)(E+p_3)+(C+p_2)(E+p_3)} \\
&\qquad \rho_a(\mathbf{L}_{-A}\chi_{rs}, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3)\rho_a(\mathbf{L}_{-B}\chi_{rs}, \mathbf{L}_{-D}\phi_2, \mathbf{L}_{-F}\phi_3).
\end{aligned} \tag{4.15}
$$

Normalizing χ_{rs} by taking the coefficient of $G^{rs}_{-1/2}\phi$ to be 1, the inverse residue of the null norm is given by

$$
\mathbf{A}^c_{rs}(h) = \lim_{h_1\to h_{r,s}} \frac{h_1 - h_{rs}}{\| \sum_M \chi^M_{rs}\mathbf{L}_{-M}\phi_1\|^2} = \frac{1}{2} \prod_{\substack{p=1-r,\ldots,r \\ q=1-s,\ldots,s \\ p+q\in 2\mathbb{Z} \\ (p,q)\neq(0,0),(r,s)}} \left[\frac{pb_{rs} + qb^{-1}_{rs}}{\sqrt{2}}\right]^{-1} \tag{4.16}
$$

The 3-point functions can be factorized as

$$
\rho_a(\mathbf{L}_{-A}\chi_{rs}, \mathbf{L}_{-B}\phi_2, \mathbf{L}_{-C}\phi_3) = \begin{cases} \rho_a(\mathbf{L}_{-A}\phi_{h_{rs}+\frac{rs}{2}}, \mathbf{L}_{-B}\phi_2, \mathbf{L}_{-C}\phi_3)\mathbf{P}^a_{rs}(h_3, h_2), & rs \in 2\mathbb{Z}, \\ (-1)^{p_1+A}\rho_{1-a}(\mathbf{L}_{-A}\phi_{h_{rs}+\frac{rs}{2}}, \mathbf{L}_{-B}\phi_2, \mathbf{L}_{-C}\phi_3)\mathbf{P}^a_{rs}(h_3, h_2), & rs \in 2\mathbb{Z}+1, \end{cases} \tag{4.17}
$$

where the two fusion polynomials are given by

$$
\begin{aligned}
\mathbf{P}^0_{rs}(h_i, h_j; c) &= \prod_{\substack{p=1-r,1-r+2,\ldots,r-1 \\ q=1-s,1-s+2,\ldots,s-1 \\ p+q-(r+s)\in 4\mathbb{Z}+2}} \frac{\lambda_i - \lambda_j + pb + qb^{-1}}{2\sqrt{2}} \frac{\lambda_i + \lambda_j + pb + qb^{-1}}{2\sqrt{2}}, \\
\mathbf{P}^1_{rs}(h_i, h_j; c) &= \prod_{\substack{p=1-r,1-r+2,\ldots,r-1 \\ q=1-s,1-s+2,\ldots,s-1 \\ p+q-(r+s)\in 4\mathbb{Z}}} \frac{\lambda_i - \lambda_j + pb + qb^{-1}}{2\sqrt{2}} \frac{\lambda_i + \lambda_j + pb + qb^{-1}}{2\sqrt{2}}.
\end{aligned} \tag{4.18}
$$

Here λ_i is defined by

$$
h_i = \frac{(b + b^{-1})^2 - \lambda_i^2}{8}. \tag{4.19}
$$

Plugging these results in the pole $\mathbf{W}^{(1)}_{a,rs}$, one then have

$$\begin{aligned}\mathbf{W}^{(1)}_{a,rs} = &-\frac{\partial c_{rs}(h_1)}{\partial h_1}\mathbf{A}^c_{rs}(h)q_1^{rs/2}\eta_1^{rs}[\mathbf{P}^a_{rs}(h_3,h_2)]^2\\ &\times\begin{cases}\mathbf{F}_a(h_1\to h_1+\frac{rs}{2}, c\to c_{rs}), \quad rs\in 2\mathbb{Z}, & rs\in 2\mathbb{Z},\\ \mathbf{F}_{1-a}(h_1\to h_1+\frac{rs}{2}, c\to c_{rs}, \eta_{2,3}\to -\eta_{2,3}), & rs\in 2\mathbb{Z}+1,\end{cases}\end{aligned}\tag{4.20}$$

where the additional shift $\eta_{2,3}\to-\eta_{2,3}$ comes from the additional cross sign factor $(-1)^{rs(C+p_2+E+p_3)}$.

Similar logic can be applied to compute the residue at $c=c_{r,s}(h_2)$ and $c=c_{r,s}(h_3)$. For that computation, one need to be careful about the sign in the factorialization of three point function. For a null state in the second slot, one have

$$\begin{aligned}&\rho_a(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\chi_{rs},\mathbf{L}_{-E}\phi_3)\\ &=\begin{cases}\mathbf{P}^a_{rs}(h_3,h_1;c_{rs})\rho_a\big(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_{h_{rs}+\frac{rs}{2}},\mathbf{L}_{-E}\phi_3\big)\,, & rs\in 2\mathbb{Z},\\ \mathbf{P}^a_{rs}(h_3,h_1;c_{rs})\rho_{1-a}\big(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_{h_{rs}+\frac{rs}{2}},\mathbf{L}_{-E}\phi_3\big)\,, & rs\in 2\mathbb{Z}+1.\end{cases}\end{aligned}\tag{4.21}$$

For a null state in the third slot,

$$\begin{aligned}&\rho_a(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-E}\chi_{rs})\\ &=\begin{cases}\mathbf{P}^a_{rs}(h_1,h_2;c_{rs})\rho_a\big(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-E}\phi_{h_{rs}+\frac{rs}{2}}\big)\,, & rs\in 2\mathbb{Z},\\ (-1)^{1+p_2}\mathbf{P}^a_{rs}(h_1,h_2;c_{rs})\rho_{1-a}\big(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-E}\phi_{h_{rs}+\frac{rs}{2}}\big)\,, & rs\in 2\mathbb{Z}+1.\end{cases}\end{aligned}\tag{4.22}$$

The additional signs will cancel out in the theta channel, giving

$$\begin{aligned}\mathbf{W}^{(2)}_{a,rs} = &-\frac{\partial c_{rs}(h_2)}{\partial h_2}\mathbf{A}^c_{rs}(h)q_2^{rs/2}\eta_2^{rs}[\mathbf{P}^a_{rs}(h_3,h_1)]^2\\ &\times\begin{cases}\mathbf{F}_a(h_2\to h_2+\frac{rs}{2}, c\to c_{rs}), \quad rs\in 2\mathbb{Z}, & rs\in 2\mathbb{Z},\\ \mathbf{F}_{1-a}(h_2\to h_2+\frac{rs}{2}, c\to c_{rs}, \eta_{3,1}\to -\eta_{3,1}), & rs\in 2\mathbb{Z}+1,\end{cases}\end{aligned}\tag{4.23}$$

$$\begin{aligned}\mathbf{W}^{(3)}_{a,rs} = &-\frac{\partial c_{rs}(h_3)}{\partial h_3}\mathbf{A}^c_{rs}(h)q_3^{rs/2}\eta_3^{rs}[\mathbf{P}^a_{rs}(h_1,h_2)]^2\\ &\times\begin{cases}\mathbf{F}_a(h_3\to h_3+\frac{rs}{2}, c\to c_{rs}), \quad rs\in 2\mathbb{Z}, & rs\in 2\mathbb{Z},\\ \mathbf{F}_{1-a}(h_3\to h_3+\frac{rs}{2}, c\to c_{rs}, \eta_{1,2}\to -\eta_{1,2}), & rs\in 2\mathbb{Z}+1.\end{cases}\end{aligned}\tag{4.24}$$

5 The Ramond blocks

Let us now consider the case with intermediate Ramond supermultiplets. The h-recursion relation of sphere 4-point Ramond block is derived in [**0810.1203**, **1012.2974**], and that of torus 1-point Ramond block is derived in [**1207.5740**]. In this section, we aim for c-recursion of arbitrary Ramond blocks.

A Ramond multiplet has 2 ground states $w^{\pm}$ that are annihilated by $L_{m>0}, G_{n>0}$ and carries L_0 eigenvalue h. They are related by acting G_0 (we use the convention of [**0810.1203**] for R-supermultiplet):

$$G_0 w^{\pm} = i\beta e^{\mp i\pi/4} w^{\mp}, \tag{5.1}$$

where β is defined by $\beta^2 = \frac{c}{24} - h$. For R-sector c-recursion relation, instead of fixing the internal weight h, we fix β of the internal primary, as doing so keeps the G_0 action on $w^{\pm}$ invariant. We take w^{+} to be Grassmann even, and hence w^{-} will be Grassmann odd. An arbitrary state in the Ramond multiplet can be then written as

$$\mathbb{L}_{-A} w^{\pm} \equiv L_{-n_1} \cdots L_{-n_k} G_{-m_1} \cdots G_{-m_\ell} w^{\pm}, \quad |A| = \sum_{i=1}^{k} n_i + \sum_{j=1}^{\ell} m_j, \quad (-1)^A \equiv (-1)^\ell, \tag{5.2}$$

where $n_1 \geq \cdots \geq n_k, m_1 > \cdots > m_\ell$. Throughout this notes, the NS-sector quantities will be written in `mathbf`, and the R-sector quantities in `mathbb`.

For a sphere with (NS, R, R) insertions, we normalize the 3-point function such that

$$\rho^{++}(\phi_1, w_2^+, w_3^+) = 1, \quad \rho^{+-}(\phi_1, w_2^+, w_3^-) = 1, \quad \rho^{-+}(\phi_1, w_2^-, w_3^+) = 1, \quad \rho^{--}(\phi_1, w_2^-, w_3^-) = 1. \tag{5.3}$$

The 3-point functions with other slot orders can be related to these basic ones by a Mobius transformation. It turns out that all $\rho^{\pm,\pm}$'s with all slot orders are normalized to 1. One can introduce [**1012.2974**]

$$\begin{aligned} \rho_0^{(\eta)}(\xi_1, \xi_2, \xi_3) &= \rho^{++}(\xi_1, \xi_2, \xi_3) + \eta \rho^{--}(\xi_1, \xi_2, \xi_3), \\ \rho_1^{(\eta)}(\xi_1, \xi_2, \xi_3) &= \rho^{+-}(\xi_1, \xi_2, \xi_3) + i\eta \rho^{-+}(\xi_1, \xi_2, \xi_3), \end{aligned} \tag{5.4}$$

for $\eta = \pm$. The subscript $0, 1$ indicates the total fermion parity, as $\rho(\mathbf{L}_{-A}\phi, \mathbb{L}_{-C} w^{\alpha}, \mathbb{L}_{-E} w^{\gamma})$ can eventually be reduced to $\rho(\phi, w^{\alpha'}, w^{\gamma'})$ with $(-1)^{|\alpha'|+|\gamma'|} = (-1)^{A+C+E+|\alpha'|+|\gamma'|}$, where $|\alpha| = 0$ for $\alpha = +$ and $|\alpha| = 1$ for $\alpha = -$, as the total fermion parity stays the same under the contour pulling used in Ward identities.

The Ramond block in the glasses channel, with spin structure specified by the three signs (η_1, η_2, η_3) and the edge 1 being NS, is defined by

$$\begin{aligned} \mathbb{F}_f^{(\eta,\eta')}(h_1, \beta_2, \beta_3, c) = & \sum_{\substack{|A|=|B|,|C|=|D|,|E|=|F| \\ (-1)^{A+C+E+|\alpha|+|\gamma|}=(-1)^f \\ (-1)^{B+D+F+|\alpha'|+|\gamma'|}=(-1)^f}} q_1^{|A|} \eta_1^{A+p_1} q_2^{|C|} \eta_2^{C+|\alpha|} q_3^{|E|} \eta_3^{E+|\gamma|} \mathbf{G}_{h_1}^{AB} \mathbb{G}_{\beta_2}^{C\alpha, D\alpha'} \mathbb{G}_{\beta_3}^{E\gamma, F\gamma'} \\ & \times (-1)^{(A+p_1)(C+|\alpha|)+(A+p_1)(E+|\gamma|)+(C+|\alpha|)(E+|\gamma|)} \\ & \times \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-C} w_2^{\alpha}, \mathbb{L}_{-E} w_3^{\gamma}) \rho_f^{(\eta')}(\mathbf{L}_{-B}\phi_1, \mathbb{L}_{-D} w_2^{\alpha'}, \mathbb{L}_{-F} w_3^{\gamma'}) \end{aligned} \tag{5.5}$$

where $\mathbb{G}$ is the inverse Gram matrix for R-multiplets (which excludes actions of G_0), and $\alpha, \alpha', \gamma, \gamma'$ are such that the total fermion parity $(-1)^{A+C+E+|\alpha|+|\gamma|} = (-1)^f$.

Let's consider the case where the w_2 multiplet develops a null descendant. This happens when

$$\beta = \beta_{rs}(c) = \frac{rb + sb^{-1}}{2\sqrt{2}}, \quad r + s \in 2\mathbb{Z} + 1 \tag{5.6}$$

and for fixed β, this becomes the following value for c:

$$\begin{aligned} x^{\mathrm{R}}_{rs}(\beta) &= \frac{4\beta^2 - rs + 2\sqrt{2}\beta\sqrt{2\beta^2 - rs}}{r^2}, \\ c^{\mathrm{R}}_{rs}(\beta) &= \frac{15}{2} + 3\left(x^{\mathrm{R}}_{rs}(\beta) + x^{\mathrm{R}}_{rs}(\beta)^{-1}\right). \end{aligned} \tag{5.7}$$

We choose $b_{rs} = \sqrt{x^{\mathrm{R}}_{rs}}$ continuously such that $\beta = \beta_{rs}(c^{\mathrm{R}}_{rs})$. Below, b_{rs} and c^{R}_{rs} refer to this branch. When this happens, the R multiplet develops a doublet of null states $\chi^{\pm}_{rs} = \sum_{M,\alpha} \chi^{M,\alpha}_{rs} G_0^{\alpha} \mathbb{L}_{-M} w^{\pm}$ happens at level $h + rs/2$, and we normalize $\chi^{\pm}_{rs}$ such that the coefficient of $L_{-1}^{rs/2} w^{\pm}$ is 1. In β variables, the weight $h_{rs} = h + rs/2$ translates to

$$\beta'_{rs} = \frac{(-1)^s}{2\sqrt{2}}\left(rb_{rs} - sb_{rs}^{-1}\right). \tag{5.8}$$

The Ramond block $\mathbb{F}^{(\eta,\eta')}_f(h_1, \beta_2, \beta_3, c)$ develops a pole when $c = c^{\mathrm{R}}_{rs}(\beta_2)$. The inverse norm for the null state is given by

$$\mathbb{A}^c_{rs} \equiv \lim_{c \to c^{\mathrm{R}}_{ra}(\beta)} \frac{\beta_{rs}(c)^2 - \beta^2}{\|\chi^+_{rs}\|^2} = 2^{rs-2}\left(rb_{rs} + sb_{rs}^{-1}\right) \prod_{\substack{p=1-r,\dots,r,q=1-s,\dots,s \\ p+q\in\mathbb{Z},(p,q)\neq(0,0)}} \left(pb_{rs} + qb_{rs}^{-1}\right)^{-1} \tag{5.9}$$

and the fusion polynomials are given by

$$\begin{aligned} \mathbb{P}^{\mathrm{R},\eta}_{rs}(h_i, \beta_j; c) = &\prod_{\substack{0\le k\le r-1,\ 0\le \ell\le s-1 \\ k+\ell\in 2\mathbb{Z}}} \frac{\lambda_i - 2\sqrt{2}\,\eta\beta_j - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}} \\ &\times \prod_{\substack{0\le k\le r-1,\ 0\le \ell\le s-1 \\ k+\ell\in 2\mathbb{Z}+1}} \frac{\lambda_i + 2\sqrt{2}\,\eta\beta_j - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}}. \end{aligned} \tag{5.10}$$

The factorization relations are summarized in appendix A. Using the relation

$$\rho^{(\eta)}_f(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}\chi^{\alpha}_{rs}, \mathbb{L}_{-E}w^{\gamma}_{\beta_j}) = \mathbb{P}^{\mathrm{R},\eta}_{rs}(h_i, \beta_j; c^{\mathrm{R}}_{rs})\rho^{(\eta)}_f(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}w^{\alpha}_{\beta'_{rs}}, \mathbb{L}_{-E}w^{\gamma}_{\beta_j}) \tag{5.11}$$

one can read off the polar value of $\mathbb{F}_f^{(\eta,\eta')}$ at $c = c^{\mathrm{R}}_{rs}(\beta_2)$:

$$
\begin{aligned}
\operatorname{Res}_{c=c^{\mathrm{R}}_{rs}(\beta_2)} \mathbb{F}_f^{(\eta,\eta')} = \frac{\partial c_{rs}(\beta_2)}{\partial \beta_2^2} \mathbb{A}^c_{rs} & \sum_{\substack{|A|=|B|,|C|=|D|,|E|=|F| \\ (-1)^{A+C+E+|\alpha|+|\gamma|}=(-1)^f \\ (-1)^{B+D+F+|\alpha'|+|\gamma'|}=(-1)^f}} q_1^{|A|}\eta_1^{A+p_1} q_2^{|C|}\eta_2^{C+|\alpha|} q_3^{|E|}\eta_3^{E+|\gamma|} \\
& \times (-1)^{(A+p_1)(C+|\alpha|)+(A+p_1)(E+|\gamma|)+(C+|\alpha|)(E+|\gamma|)} \mathbf{G}^{AB}_{h_1} \mathbb{G}^{C\alpha,D\alpha'}_{\beta'_{rs}} \mathbb{G}^{E\gamma,F\gamma'}_{\beta_3} \\
& \times \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-C}\chi^\alpha_{rs}, \mathbb{L}_{-E} w_3^\gamma) \rho_f^{(\eta')}(\mathbf{L}_{-B}\phi_1, \mathbb{L}_{-D}\chi^\alpha_{rs}, \mathbb{L}_{-F} w_3^{\gamma'}),
\end{aligned}
\tag{5.12}
$$

and hence

$$
\begin{aligned}
\operatorname{Res}_{c=c^{\mathrm{R}}_{rs}(\beta_2)} \mathbb{F}_f^{(\eta,\eta')} = {} & \frac{\partial c_{rs}(\beta_2)}{\partial \beta_2^2} \mathbb{A}^c_{rs} \mathbb{P}^{\mathrm{R},\eta}_{rs}(h_1, \beta_3; c^{\mathrm{R}}_{rs}(\beta_2)) \mathbb{P}^{\mathrm{R},\eta'}_{rs}(h_1, \beta_3; c^{\mathrm{R}}_{rs}(\beta_2)) \\
& \times \mathbb{F}_f^{(\eta,\eta')}(\beta_2 \to \beta'_{rs}, c \to c^{\mathrm{R}}_{rs}(\beta_2))
\end{aligned}
\tag{5.13}
$$

where we used the fact that $\chi^{M,\alpha}_{rs} G^\alpha_0 \mathbb{L}_{-M}$ is Grassmann even, which ensures that the total fermion parity f doesn't get shifted and there are no additional signs η_2. The residue at $c = c^{\mathrm{R}}_{rs}(\beta_3)$ can be worked out analogously.

The Ramond also develops poles when the NS supermultiplet contains a null state. The relavant fusion polynomials are

$$
\begin{aligned}
\mathbb{P}^{\mathrm{NS},\eta}_{rs}(\beta_2, \beta_3; c) = & \prod_{\substack{0\le k\le r-1,\ 0\le \ell\le s-1 \\ k+\ell\in 2\mathbb{Z}}} \frac{2\sqrt{2}(\beta_3 - \eta\beta_2) - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}} \\
\times & \prod_{\substack{0\le k\le r-1,\ 0\le \ell\le s-1 \\ k+\ell\in 2\mathbb{Z}+1}} \frac{2\sqrt{2}(\beta_3 + \eta\beta_2) - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}},
\end{aligned}
\tag{5.14}
$$

and the factorization relation is given by

$$
\begin{aligned}
& \rho_f^{(\eta)}(\mathbf{L}_{-A}\chi_{rs}, \mathbb{L}_{-C} w^\alpha_{\beta_2}, \mathbb{L}_{-E} w^\gamma_{\beta_3}) \\
& \quad = \mathbb{P}^{\mathrm{NS},(-1)^{p_1}\eta}_{rs}(\beta_2, \beta_3; c_{rs}) \begin{cases} \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-C} w^\alpha_{\beta_2}, \mathbb{L}_{-E} w^\gamma_{\beta_3}), & rs \in 2\mathbb{Z}, \\ (-1)^{p_1+|\gamma|+E} e^{i(-1)^f\pi/4} \rho_{1-f}^{(-\eta)}(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-C} w^\alpha_{\beta_2}, \mathbb{L}_{-E} w^\gamma_{\beta_3}), & rs \in 2\mathbb{Z}+1. \end{cases}
\end{aligned}
\tag{5.15}
$$

From these, one can derive the residue at $c = c_{rs}(h_1)$:

$$
\begin{aligned}
\operatorname{Res}_{c=c_{rs}(h_1)} \mathbb{F}_f^{(\eta,\eta')} = {} & -\frac{\partial c_{rs}(h_1)}{\partial h_1} \mathbf{A}^c_{rs}(h_1) q_1^{rs/2} \mathbb{P}^{\mathrm{NS},(-1)^{p_1}\eta}_{rs}(\beta_2, \beta_3; c_{rs}(h_1)) \mathbb{P}^{\mathrm{NS},(-1)^{p_1}\eta'}_{rs}(\beta_2, \beta_3; c_{rs}(h_1)) \\
& \times \begin{cases} \mathbb{F}_f^{(\eta,\eta')}\left(h_1 \to h_1 + \frac{rs}{2}, c \to c_{rs}(h_1)\right), & rs \in 2\mathbb{Z}, \\ i(-1)^f \eta_1 \mathbb{F}_{1-f}^{(-\eta,-\eta')}\left(h_1 \to h_1 + \frac{rs}{2}, c \to c_{rs}(h_1), \eta_2 \to -\eta_2, \eta_3 \to -\eta_3\right), & rs \in 2\mathbb{Z}+1. \end{cases}
\end{aligned}
\tag{5.16}
$$

We now have computed all the polar part in the following recursion relation

$$\begin{aligned}\mathbb{F}_f^{(\eta,\eta')}(h_1,\beta_2,\beta_3,c) &= \mathbb{F}_f^{(\eta,\eta')}(h_1,\beta_2,\beta_3,c\to\infty) \\ &\quad + \sum_{r+s\in 2\mathbb{Z}} \frac{\mathrm{Res}_{c=c_{rs}(h_1)}\,\mathbb{F}_f^{(\eta,\eta')}}{c-c_{rs}(h_1)} + \sum_{i=2,3}\sum_{r+s\in 2\mathbb{Z}+1} \frac{\mathrm{Res}_{c=c^{\mathrm{R}}_{rs}(\beta_i)}\,\mathbb{F}_f^{(\eta,\eta')}}{c-c^{\mathrm{R}}_{rs}(\beta_i)}\end{aligned} \tag{5.17}$$

The next ingredient of the c-recursion relation of the Ramond blocks is the large-c limit, which we will compute in the next section.

6 The large-c Ramond block

In the $c\to\infty$ limit, we fix h of the NS supermultiplet, and fix β of the R supermultiplet, which implies that $L_0 = \frac{c}{24} - \beta^2 + |A|$ is of order $\mathcal{O}(c)$ for a Ramond state $\mathbb{L}_{-A}w_\beta^\pm$.

We first analyze the c-scaling behavior of the Gram matrix element $\langle w_\beta^\alpha|\mathbb{L}_{-A}^\dagger\mathbb{L}_{-B}|w_\beta^\gamma\rangle$, where there are no G_0's in the lowering operators $\mathbb{L}_{-A}$. The SCA in the R-sector reads

$$\begin{aligned}[L_m,L_n] &= (m-n)L_{m+n} + \frac{c}{12}\left(m^3-m\right)\delta_{m+n,0}, \\ [L_m,G_n] &= \left(\frac{m}{2}-n\right)G_{m+n}, \\ \{G_m,G_n\} &= 2L_{m+n} + \frac{c}{3}\left(m^2-\frac{1}{4}\right)\delta_{m+n,0}.\end{aligned} \tag{6.1}$$

Therefore, in the large-c limit, the algebra becomes

$$[L_m,L_{-n}] = \frac{c}{12}m^3\delta_{m,n}, \quad \{G_m,G_{-n}\} = \frac{c}{3}m^2\delta_{m,n}, \quad [L_m,G_{-n}] = 0. \tag{6.2}$$

The leading term in the Gram matrix then comes from pairwise contractions between the generators. Let $\hat{A}$ be the total number of generators in $\mathbb{L}_{-A}$, then $\langle w_\beta^\alpha|\mathbb{L}_{-A}^\dagger\mathbb{L}_{-B}|w_\beta^\gamma\rangle$ is of order $\mathcal{O}(c^{\hat{A}})$ if $(A,\alpha)=(B,\gamma)$, at most $\mathcal{O}(c^{\hat{A}-1})$ if $\hat{A}=\hat{B},(A,\alpha)\neq(B,\gamma)$, and at most $\mathcal{O}(c^{\min(\hat{A},\hat{B})})$ if $\hat{A}\neq\hat{B}$. One can then apply Gram-Schmidt to $\mathbb{L}_{-A}w_\beta^\pm$ to obtain a orthonormal basis $\mathcal{L}_{-A}w_\beta^\pm$, and

$$\|\mathcal{L}_{-A}w_\beta^\pm\|^2 \sim \mathcal{O}(c^{\hat{A}}). \tag{6.3}$$

The next ingredient is the scaling behavior of the 3-point function $\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1,\mathbb{L}_{-B}w_2^\alpha,\mathbb{L}_{-C}w_3^\gamma)$. One can use conformal Ward identities to convert a L_{-n},G_{-r} insertion in one slot to another slot. For a L_{-n} insertion,

$$\begin{aligned}
&\rho_f^{(\eta)}\left(L_{-n}\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma\right) \\
&\qquad = \rho_f^{(\eta)}\left(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, L_n\mathbb{L}_{-C}w_3^\gamma\right) + \sum_{p=-1}^{n}\binom{n+1}{p+1}\rho_f^{(\eta)}\left(\mathbf{L}_{-A}\phi_1, L_p\mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma\right), \\
&\rho_f^{(\eta)}\left(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, L_{-n}\mathbb{L}_{-C}w_3^\gamma\right) \\
&\qquad = \rho_f^{(\eta)}\left(L_n\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma\right) - \sum_{p=0}^{\infty}\binom{1-n}{p}\rho_f^{(\eta)}\left(\mathbf{L}_{-A}\phi_1, L_{p-1}\mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma\right), \\
&\rho_f^{(\eta)}\left(\mathbf{L}_{-A}\phi_1, L_{-n}\mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma\right) = \sum_{p=0}^{\infty}\binom{n-2+p}{n-2}\rho_f^{(\eta)}\left(L_{n+p}\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma\right) \\
&\qquad + (-1)^n\sum_{p=0}^{\infty}\binom{n-2+p}{n-2}\rho_f^{(\eta)}\left(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, L_{p-1}\mathbb{L}_{-C}w_3^\gamma\right).
\end{aligned} \tag{6.4}$$

This imposes a problem to our infinite-c limit. Each conformal Ward identity contains a L_0 term acting on a Ramond multiplet on the RHS. In our fixed β large-c limit, this will contribute to a term of order $\mathcal{O}(c)$. Therefore, for $\mathbf{L}_{-A}, \mathbb{L}_{-A}$ made from purely L_{-n}'s,

$$\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma) \sim \mathcal{O}(c^{\hat{A}+\hat{B}+\hat{C}}) \tag{6.5}$$

and the contribution of this term to the large-c conformal block will scale like $\mathcal{O}(c^{\hat{A}+\hat{B}+\hat{C}})$. Namely, the large-$c$, fixed-$\beta$ limit of the Ramond block is not well-defined.

7 NS blocks from double Virasoro

We now turn to another approach of computing the superconformal blocks. The idea is: Given the $\mathcal{N} = 1$ SCA with shift $\nu \in \left\{0, \frac{1}{2}\right\}$ in the fermionic level, one can construct an enlarged algebra $\mathsf{SCA}_\nu \otimes \mathsf{F}_\nu$, where F_ν a auxiliary free fermion algebra with the same level shift ν. $\mathsf{SCA}_\nu \otimes \mathsf{F}_\nu$ admits two copy of virasoro algebra as it's subalgebra $\mathsf{Vir} \oplus \mathsf{Vir} \subset \mathsf{SCA}_\nu \otimes \mathsf{F}_\nu$ [**1312.4520**, **1608.02568**]. Using the branching rules of this subalgebra reduction, one can decompose the superconformal block into ordinary Virasoro blocks.

The enlarged algebra $\mathsf{SCA}_\nu \otimes \mathsf{F}_\nu$ is given by:

$$
\begin{aligned}
&[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}\left(m^3 - m\right)\delta_{m+n,0}, \\
&[L_m, G_r] = \left(\frac{m}{2} - r\right) G_{m+r}, \\
&\{G_r, G_s\} = 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}, \\
&\{\psi_r, \psi_s\} = \delta_{r+s,0}, \quad \{\psi_r, G_s\} = [\psi_r, L_n] = 0.
\end{aligned}
\tag{7.1}
$$

where the levels $m, n \in \mathbb{Z}$, $r, s \in \mathbb{Z} + \nu$. The reality conditions are

$$
\psi_r^\dagger = -\psi_{-r}, \quad G_r^\dagger = G_{-r}, \quad L_n^\dagger = L_{-n}. \tag{7.2}
$$

We also introduce the following operators

$$
L_n^{\mathrm{F}} = \frac{1}{2}\sum_{r\in\mathbb{Z}+\nu} r : \psi_{n-r}\psi_r : + \frac{1-2\nu}{16}\delta_{n,0}, \quad U_n = \sum_{r\in\mathbb{Z}+\nu} \psi_{n-r}G_r. \tag{7.3}
$$

Define b by the relation

$$
c = \frac{3}{2} + 3\left(b + b^{-1}\right)^2. \tag{7.4}
$$

For $b \neq 1$, one can then define

$$
\begin{aligned}
L_n^{(1)} &= \frac{b^{-1}}{b^{-1} - b} L_n - \frac{b^{-1} + 2b}{b^{-1} - b} L_n^{\mathrm{F}} + \frac{1}{b^{-1} - b} U_n, \\
L_n^{(2)} &= \frac{b}{b - b^{-1}} L_n - \frac{b + 2b^{-1}}{b - b^{-1}} L_n^{\mathrm{F}} + \frac{1}{b - b^{-1}} U_n,
\end{aligned}
\tag{7.5}
$$

which satisfies $L_n^{(1)} + L_n^{(2)} = L_n + L_n^{\mathrm{F}}$. Note that in order to have $(L_n^{(i)})^\dagger = L_{-n}^{(i)}$, we need to take the BPZ conjugate of ψ_r to be $(\psi_r)^\dagger = -\psi_{-r}$. One can also prove that these operators are the generators of two commuting copies of the Virasoro algebra

$$
[L_m^{(i)}, L_n^{(j)}] = \left[(m-n)L_{m+n}^{(i)} + \frac{c^{(i)}}{12}(m^3 - m)\delta_{m+n,0}\right]\delta_{ij}, \tag{7.6}
$$

and the two central charges are given by

$$
c^{(i)} = 1 + 6\left(b^{(i)} + \frac{1}{b^{(i)}}\right)^2, \quad (b^{(1)})^2 = \frac{2b^2}{1-b^2}, \quad (b^{(2)})^{-2} = \frac{2}{b^2 - 1}. \tag{7.7}
$$

This proves the subalgebra relation $\mathsf{Vir} \oplus \mathsf{Vir} \subset \mathsf{SCA}_\nu \otimes \mathsf{F}_\nu$.

Under the subalgebra reduction $\mathsf{Vir} \oplus \mathsf{Vir} \subset \mathsf{SCA}_\nu \otimes \mathsf{F}_\nu$, a highest-weight representation of $\mathsf{SCA}_\nu \otimes \mathsf{F}_\nu$ can be decomposed into irreps of $\mathsf{Vir} \oplus \mathsf{Vir}$. This decomposition is worked out in [**1608.02568**]. We now discuss this decomposition in both NS and R sectors.

NS Sector

As always, let $\mathcal{V}_{\mathsf{SCA}_{1/2}}(h)$ denote the NS Verma module generated by an highest-weight state ϕ of conformal weight h. Let $\mathcal{V}_{F_{1/2}}$ denote the Fock representation of $F_{1/2}$. The representation of interest is $\mathcal{M}_{\mathrm{NS}}(h) := \mathcal{V}_{F_{1/2}} \otimes \mathcal{V}_{\mathsf{SCA}_{1/2}}(h)$. It can be generated by acting arbitrary $\psi_{-r}, L_{-n}, G_{-r}$ on $\mathbf{1}_{\mathsf{F}} \otimes \phi$.

We further define the "Liouville momentum" P by

$$h = \frac{1}{2}\left(\frac{1}{4}Q^2 - P^2\right), \tag{7.8}$$

where $Q = b + b^{-1}$. Denoting the Virasoro Verma module with lowest weight h by $\mathcal{V}_c(h)$, it's proven that [**1111.2803**]

$$\mathcal{M}_{\mathrm{NS}}(h) \cong \bigoplus_{n \in \frac{1}{2}\mathbb{Z}} \mathcal{V}_{c^{(1)}}(h_n^{(1)}) \otimes \mathcal{V}_{c^{(2)}}(h_n^{(2)}) \tag{7.9}$$

for $P \notin \frac{1}{2}(b\mathbb{Z} + b^{-1}\mathbb{Z})$. The weights $h_n^{(i)}$'s are given by $h_n^{(i)} = \frac{1}{4}(Q^{(i)})^2 - (P_n^{(i)})^2$, where

$$P_n^{(1)} = \frac{P}{\sqrt{2-2b^2}} + nb^{(1)}, \quad P_n^{(2)} = \frac{P}{\sqrt{2-2b^{-2}}} + n(b^{(2)})^{-1}. \tag{7.10}$$

It satisfies $h_n^{(1)} + h_n^{(2)} = h + 2n^2$.

We will denote the highest weight vector in $\mathcal{V}_{c^{(1)}}(h_n^{(1)}) \otimes \mathcal{V}_{c^{(2)}}(h_n^{(2)})$ by v_n. In our convention where ϕ is Grassmann even, v_n has Grassmann parity $2n$. The Liouville momentum P gives identical $\mathcal{V}_{\mathsf{SCA}_{1/2}}$, and a reflection $(n, P) \mapsto (-n, -P)$ gives the same $\mathsf{Vir} \oplus \mathsf{Vir}$ module. We can then identify $v_n(P) = v_{-n}(-P)$ and work primarily with $P > 0$. For multiplets with $h > Q^2/8$, we analytic continue all the formula from positive real P to purely imaginary $P = ip$ with $p > 0$.

For our application, we will always use the following bilinear pairing for $\mathcal{M}_{\mathrm{NS}}(h)$

$$\left\langle\!\left\langle \mathbf{\Psi}_{-\mathsf{A}}\mathbf{1}_F \otimes \mathbf{L}_{-A}\phi \,\middle|\, \mathbf{\Psi}_{-\mathsf{B}}\mathbf{1}_F \otimes \mathbf{L}_{-B}\phi \right\rangle' = \left\langle\!\left\langle \mathbf{\Psi}_{-\mathsf{A}}\mathbf{1}_F \,\middle|\, \Psi_{-\mathbf{B}}\mathbf{1}_F \right\rangle \times \left\langle\!\left\langle \mathbf{L}_{-A}\phi \,\middle|\, \mathbf{L}_{-B}\phi \right\rangle'. \tag{7.11}$$

where we use $A + B = 0$ mod2 for the matrix element to be non-vanishing. This bilinear pairing has the property that

$$\left\langle\!\left\langle \mathbf{L}^{(1)}_{-M}\nu_1 \otimes \mathbf{L}^{(2)}_{-N}\nu_2 \,\middle|\, \mathbf{L}^{(1)}_{-M'} \otimes \nu_1 \mathbf{L}^{(2)}_{-N'} \otimes \nu_2 \right\rangle' = \left\langle\!\left\langle \mathbf{L}^{(1)}_{-M}\nu_1 \,\middle|\, \mathbf{L}^{(1)}_{-M'}\nu_1 \right\rangle \left\langle\!\left\langle \mathbf{L}^{(2)}_{-N}\nu_2 \,\middle|\, \mathbf{L}^{(2)}_{-N'}\nu_2 \right\rangle'. \tag{7.12}$$

It should be distinguished from the standard bilinear pairing on $\mathcal{M}_{\mathrm{NS}}(h)$ in the theory of chiral fermion coupled to superconformal field theory, where one need an additional crossing sign

$$\left\langle\!\left\langle \mathbf{\Psi}_{-\mathsf{A}}\mathbf{1}_F \otimes \mathbf{L}_{-A}\phi \,\middle|\, \mathbf{\Psi}_{-\mathsf{B}}\mathbf{1}_F \otimes \mathbf{L}_{-B}\phi \right\rangle = (-1)^{\mathsf{B}(A+p_\phi)} \left\langle\!\left\langle \mathbf{\Psi}_{-\mathsf{A}}\mathbf{1}_F \,\middle|\, \Psi_{-\mathbf{B}}\mathbf{1}_F \right\rangle \times \left\langle\!\left\langle \mathbf{L}_{-A}\phi \,\middle|\, \mathbf{L}_{-B}\phi \right\rangle. \tag{7.13}$$

R Sector

Let $\mathcal{V}_{\mathsf{SCA}_0}(\beta)$ denote the R Verma module generated by an highest-weight state w_β^+ of conformal weight $h = \frac{c}{24} - \beta^2$. Two degenerate ground states will be denoted as $w_\beta^\pm$. Let $\mathcal{V}_{F_0}$ denote the Fock representation of F_0. Two ground states in $\mathcal{V}_{F_0}$ will be denoted as $u^\pm$, with $\psi_0 u^\pm = 2^{-1/2}u^\mp$.

The representation of interest is $\mathcal{M}_{\mathrm{R}}(\beta) := \mathcal{V}_{F_0} \otimes \mathcal{V}_{\mathsf{SCA}_0}(\beta)$. It contains 4 ground states given by $u^\pm \otimes w_\beta^\pm$. $\mathcal{M}_{\mathrm{R}}(\beta)$ can be generated by acting lowering operators on the 4 ground states $u^\pm \otimes w_\beta^\pm$. The Liouville momentum of $w_\beta^\pm$ is defined by

$$h = \frac{c}{24} - \beta^2 = \frac{1}{16} + \frac{1}{2}\left(\frac{1}{4}Q^2 - P^2\right), \quad P^2 = 2\beta^2. \tag{7.14}$$

For $P \notin \frac{1}{2}(b\mathbb{Z} + b^{-1}\mathbb{Z})$, $\mathcal{M}_{\mathrm{R}}(\beta)$ can be decomposed into [**1608.02568**]

$$\mathcal{M}_{\mathrm{R}}(\beta) \cong \bigoplus_{\epsilon=0,1} \bigoplus_{n\in\frac{1}{2}\mathbb{Z}+\frac{1}{4}} \mathcal{V}_{c^{(1)}}(h_n^{(1)}) \otimes \mathcal{V}_{c^{(2)}}(h_n^{(2)}), \tag{7.15}$$

where

$$h_n^{(i)} = \frac{1}{4}(Q^{(i)})^2 - (P_n^{(i)})^2, \quad P_n^{(1)} = \frac{P}{\sqrt{2-2b^2}} + nb^{(1)}, \quad P_n^{(2)} = \frac{P}{\sqrt{2-2b^{-2}}} + n(b^{(2)})^{-1}. \tag{7.16}$$

It satisfies $h_n^{(1)} + h_n^{(2)} = h + 2n^2 - \frac{1}{16}$.

The direct sum over ϵ indicates that there are two copies of each $\mathcal{V}_{c^{(1)}}(h_n^{(1)}) \otimes \mathcal{V}_{c^{(2)}}(h_n^{(2)})$, and ϵ denotes the total fermion parity of the lowest weight state (including the ψ parity). One can directly check that the characters of the two sides of (7.9), (7.15) agrees via the Jacobi triple product identity. The ground state of $\mathcal{V}_{c^{(1)}}(h_n^{(1)}) \otimes \mathcal{V}_{c^{(2)}}(h_n^{(2)})$ will be denoted by $v_n^\pm$. We use the convention that v_n^α has Grassmann parity α.

Decomposition of all-NS block

Let's inspect on how the superconformal blocks $\mathbf{F}_a(h_1,h_2,h_3,c)$, $\mathbb{F}_f^{(\eta,\eta')}(h_1,\beta_2,\beta_3,c)$ decompose under $\mathsf{Vir} \oplus \mathsf{Vir} \subset \mathsf{SCA}_\nu \otimes \mathsf{F}$. We first define the all-NS $\mathsf{SCA}_\nu \otimes \mathsf{F}$ block in the theta channel as

$$\begin{aligned}\widehat{\mathbf{F}}_a(h_1,h_2,h_3,c) = &\sum_{\substack{|A|=|A'|,|B|=|B'|,|C|=|C'|,\mathsf{A},\mathsf{B},\mathsf{C} \\ (-1)^{A+B+C}(-1)^{\mathsf{A}+\mathsf{B}+\mathsf{C}}=(-1)^a}} q_1^{|A|}\eta_1^{A+p_1} q_2^{|B|}\eta_2^{B+p_2} q_3^{|C|}\eta_3^{C+p_3} \mathbf{G}_{h_1}^{AA'}\mathbf{G}_{h_2}^{BB'}\mathbf{G}_{h_3}^{CC'} \\ &\times (-1)^{(A+\mathsf{A}+p_1)(B+\mathsf{B}+p_2)+(A+\mathsf{A}+p_1)(C+\mathsf{C}+p_3)+(B+\mathsf{B}+p_2)(C+\mathsf{C}+p_3)}(-1)^{\mathsf{A}+\mathsf{B}+\mathsf{C}} q_1^{|\mathsf{A}|}\eta_1^{\mathsf{A}} q_2^{|\mathsf{B}|}\eta_2^{\mathsf{B}} q_3^{|\mathsf{C}|}\eta_3^{\mathsf{C}} \\ &\times \hat{\rho}_a(\Psi_{-\mathsf{A}}\mathbf{L}_{-A}\phi_1, \Psi_{-\mathsf{B}}\mathbf{L}_{-B}\phi_2, \Psi_{-\mathsf{C}}\mathbf{L}_{-C}\phi_3)\hat{\rho}_a(\Psi_{-\mathsf{A}}\mathbf{L}_{-A'}\phi_1, \Psi_{-\mathsf{B}}\mathbf{L}_{-B'}\phi_2, \Psi_{-\mathsf{C}}\mathbf{L}_{-C'}\phi_3),\end{aligned} \tag{7.17}$$

where $\Psi_{-\mathsf{A}}v$ stands for $\Psi_{-\mathsf{A}}v \equiv \psi_{-r_1}\cdots\psi_{-r_m}v$, $|\mathsf{A}|$ stands for the total level of fermion lowering operators, and $(-1)^{\mathsf{A}}$ is the total ψ parity. We have used the fact that $\left\langle\!\left\langle \mathbf{L}_{-A}v \mid \Psi^{\dagger}_{-\mathsf{A}}\Psi_{-\mathsf{B}}\middle|\mathbf{L}_{-B}v\right\rangle\right.$ is the matrix $(-1)^{\mathsf{A}}\delta_{\mathsf{A},\mathsf{B}}\mathbf{G}_{AB}$, which give rises to the sign $(-1)^{\mathsf{A}+\mathsf{B}+\mathsf{C}}$. The 3-point function $\hat{\rho}^{L/R}$ is labeled by the total parity, including the fermionic one.

The three point function could factorize into(this should be viewed as the definition of $\widehat{\rho}_a$, so that the latter double virasoro factorization works)

$$\begin{aligned}&\widehat{\rho}_a(\Psi_{-\mathsf{A}}\mathbf{L}_{-A}\phi_1,\,\Psi_{-\mathsf{B}}\mathbf{L}_{-B}\phi_2,\,\Psi_{-\mathsf{C}}\mathbf{L}_{-C}\phi_3)\\&\quad=(-1)^{\,B\mathsf{C}+p_1A+p_2C}\,\rho_{\mathsf{F}}(\Psi_{-\mathsf{A}}\mathbf{1},\,\Psi_{-\mathsf{B}}\mathbf{1},\,\Psi_{-\mathsf{C}}\mathbf{1})\,\rho_a(\mathbf{L}_{-A}\phi_1,\,\mathbf{L}_{-B}\phi_2,\,\mathbf{L}_{-C}\phi_3)\,.\end{aligned}\tag{7.18}$$

where $\rho^{\mathsf{F}}(u_1,u_2,u_3)$ is the normalized 3-point function in the free fermion theory. Importantly, ρ^{F} is only nonzero if $\mathsf{A}+\mathsf{B}+\mathsf{C}\equiv 0 \bmod 2$. If we define the free fermion block as

$$\mathbf{F}^{\mathsf{F}}(q_i,\eta_i)=\sum_{\substack{\mathsf{A},\mathsf{B},\mathsf{C}\\(-1)^{\mathsf{A}+\mathsf{B}+\mathsf{C}}=1}}(-1)^{\mathsf{AB}+\mathsf{AC}+\mathsf{BC}}q_1^{|\mathsf{A}|}\eta_1^{\mathsf{A}}q_2^{|\mathsf{B}|}\eta_2^{\mathsf{B}}q_3^{|\mathsf{C}|}\eta_3^{\mathsf{C}}\left[\rho^{\mathsf{F}}(\Psi_{-\mathsf{A}}\mathbf{1}_{\mathsf{F}},\Psi_{-\mathsf{B}}\mathbf{1}_{\mathsf{F}},\Psi_{-\mathsf{C}}\mathbf{1}_{\mathsf{F}})\right]^2,\tag{7.19}$$

then the all-NS $\mathsf{SCA}_\nu\otimes\mathsf{F}$ block can be factorized into

$$\widehat{\mathbf{F}}_a(h_1,h_2,h_3,c;q_i,\eta_i)=\mathbf{F}_{\mathsf{F}}(q_i,\eta_i)\star\mathbf{F}_a(h_1,h_2,h_3,c;q_i,\eta_i),\tag{7.20}$$

where the symbol $\star$ stands for convolution: Denote the coefficient of $q_1^{n_1/2}q_2^{n_2/2}q_3^{n_3/2}$ in the q-series $\mathbf{F}_a$ by $\mathbf{F}_a[q_1^{n_1/2}q_2^{n_2/2}q_3^{n_3/2}]$, this equation can be written as

$$\begin{aligned}&\widehat{\mathbf{F}}_a(h_1,h_2,h_3,c)[q_1^{n_1/2}q_2^{n_2/2}q_3^{n_3/2}]\\&\quad=\sum_{r_i+s_i=n_i}(-1)^{\sum_{i<j}(r_i(s_j+p_j)+(s_i+p_i)r_j)}\mathbf{F}_{\mathsf{F}}[q_1^{r_1/2}q_2^{r_2/2}q_3^{r_3/2}]\mathbf{F}_a(h_1,h_2,h_3,c)[q_1^{s_1/2}q_2^{s_2/2}q_3^{s_3/2}]\end{aligned}\tag{7.21}$$

Using the decomposition of the representation $\mathcal{M}_{\mathrm{NS}}$ under $\mathsf{Vir}\oplus\mathsf{Vir}\subset\mathsf{SCA}_\nu\otimes\mathsf{F}$, we can also express $\widehat{\mathbf{F}}_a(h_1,h_2,h_3,c)$ in terms of Virasoro blocks. The three point function will factorize into three point function of double virasoro as

$$\begin{aligned}&\widehat{\rho}_a\Big(L^{(1)}_{-M_1}L^{(2)}_{-N_1}v_1^{p_1},L^{(1)}_{-M_2}L^{(2)}_{-N_2}v_2^{p_2},L^{(1)}_{-M_3}L^{(2)}_{-N_3}v_3^{p_3}\Big)\\&\quad=\widehat{\rho}_a(v_1^{p_1},v_2^{p_2},v_3^{p_3})\,\rho^{(1)}(L_{-M_1}\nu_1,L_{-M_2}\nu_2,L_{-M_3}\nu_3)\,\rho^{(2)}(L_{-N_1}\nu_1,L_{-N_2}\nu_2,L_{-N_3}\nu_3).\end{aligned}\tag{7.22}$$

The all-NS $\mathsf{SCA}_\nu \otimes \mathsf{F}$ block can be expressed as

$$\begin{aligned}
\widehat{\mathbf{F}}_a(h_1,h_2,h_3,c) = &\sum_{\substack{|A|=|A'|,|B|=|B'|,|C|=|C'|\\ |D|=|D'|,|E|=|E'|,|F|=|F'|\\ n_1,n_2,n_3\in\frac{1}{2}\mathbb{Z}\\ (-1)^{2(n_1+n_2+n_3)}=(-1)^a}} q_1^{2n_1^2+|A|+|B|} q_2^{2n_2^2+|C|+|D|} q_3^{2n_3^2+|E|+|F|} \eta_1^{2n_1+p_1}\eta_2^{2n_2+p_2}\eta_3^{2n_3+p_3}\\
&\times (-1)^{(2n_1+p_1)(2n_2+p_2)+(2n_1+p_1)(2n_3+p_3)+(2n_2+p_2)(2n_3+p_3)}\\
&\frac{(G_{h^{(1)}_{1,n_1}})^{AA'}(G_{h^{(2)}_{1,n_1}})^{BB'}(G_{h^{(1)}_{2,n_2}})^{CC'}(G_{h^{(2)}_{2,n_2}})^{DD'}(G_{h^{(1)}_{3,n_3}})^{EE'}(G_{h^{(2)}_{3,n_3}})^{FF'}}{\|v^{p_1}_{1,n_1}\|^2\|v^{p_2}_{2,n_2}\|^2\|v^{p_3}_{3,n_3}\|^2}\\
&\times \hat\rho_a(L^{(1)}_{-A}L^{(2)}_{-B}v^{p_1}_{1,n_1}, L^{(1)}_{-C}L^{(2)}_{-D}v^{p_2}_{2,n_2}, L^{(1)}_{-E}L^{(2)}_{-F}v^{p_3}_{3,n_3})\hat\rho_a(L^{(1)}_{-A'}L^{(2)}_{-B'}v^{p_1}_{1,n_1}, L^{(1)}_{-C'}L^{(2)}_{-D'}v^{p_2}_{2,n_2}, L^{(1)}_{-E'}L^{(2)}_{-F'}v^{p_3}_{3,n_3}),
\end{aligned} \tag{7.23}$$

where we have used the fact that $h^{(1)}_n + h^{(2)}_n - h = 2n^2$ and that v_n has Grassmann parity $2n$. G_h^{AB} is the Virasoro inverse Gram matrix. The resolution of $(-1)^{p_{\mathsf{SCA}}\cdot p_{\mathsf{F}}}$ leads to the factorialization of the Gram matrix.

The 3-point function $\hat\rho_a(L^{(1)}_{-A}L^{(2)}_{-B}v^{p_1}_{1,n_1}, L^{(1)}_{-C}L^{(2)}_{-D}v^{p_2}_{2,n_2}, L^{(1)}_{-E}L^{(2)}_{-F}v^{p_3}_{3,n_3})$ can be reduced to those of Virasoro primaries $\hat\rho_a(v^{p_1}_{1,n_1}, v^{p_2}_{2,n_2}, v^{p_3}_{3,n_3})$ by conformal Ward identity. The two copies of Virasoro can be reduced independently, which leads to the factorization formula

$$\begin{aligned}
&\hat\rho_a(L^{(1)}_{-A}L^{(2)}_{-B}v^{p_1}_{1,n_1}, L^{(1)}_{-C}L^{(2)}_{-D}v^{p_2}_{2,n_2}, L^{(1)}_{-E}L^{(2)}_{-F}v^{p_3}_{3,n_3})\\
&\quad = \hat\rho_a(v^{p_1}_{1,n_1}, v^{p_2}_{2,n_2}, v^{p_3}_{3,n_3})\rho(L_{-A}\nu_{h^{(1)}_{1,n_1}}, L_{-C}\nu_{h^{(1)}_{2,n_2}}, L_{-E}\nu_{h^{(1)}_{3,n_3}})\rho(L_{-B}\nu_{h^{(2)}_{1,n_1}}, L_{-D}\nu_{h^{(2)}_{2,n_2}}, L_{-F}\nu_{h^{(2)}_{3,n_3}}),
\end{aligned} \tag{7.24}$$

where $\rho(L_{-A}\nu_1, L_{-B}\nu_2, L_{-C}\nu_3)$ is defined in section 1. If we define the branching coefficient

$$B_a(h_1,h_2,h_3;n_1,n_2,n_3) = \frac{\hat\rho_a(v^{p_1}_{1,n_1}, v^{p_2}_{2,n_2}, v^{p_3}_{3,n_3})}{\|v^{p_1}_{1,n_1}\|\|v^{p_2}_{2,n_2}\|\|v^{p_3}_{3,n_3}\|}, \tag{7.25}$$

the $\mathsf{SCA}_\nu \otimes \mathsf{F}$ block can be decomposed into

$$\begin{aligned}
\widehat{\mathbf{F}}_a(h_1,h_2,h_3,c) = &\sum_{\substack{n_1,n_2,n_3\in\frac{1}{2}\mathbb{Z}\\ (-1)^{2(n_1+n_2+n_3)}=(-1)^a}} q_1^{2n_1^2}q_2^{2n_2^2}q_3^{2n_3^2}\eta_1^{2n_1+p_1}\eta_2^{2n_2+p_2}\eta_3^{2n_3+p_3}\\
&\times (-1)^{(2n_1+p_1)(2n_2+p_2)+(2n_1+p_1)(2n_3+p_3)+(2n_2+p_2)(2n_3+p_3)}\\
&\times [B_a(h_1,h_2,h_3;n_1,n_2,n_3)]^2 F(h^{(1)}_{1,n_1}, h^{(1)}_{2,n_2}, h^{(1)}_{3,n_3}, c^{(1)}) F(h^{(2)}_{1,n_1}, h^{(2)}_{2,n_2}, h^{(2)}_{3,n_3}, c^{(2)}),
\end{aligned} \tag{7.26}$$

where $F(h_1,h_2,h_3,c)$ is the Virasoro conformal block, which can be computed efficiently using the c-relation relation in section 1. Since the fermion block $\mathbf{F}_{\mathsf{F}}$ will always have the identity term 1, one can find $\mathbf{F}_{\mathsf{F}}^{-1}$ as a formal power series in q, such that $\mathbf{F}_{\mathsf{F}} \star \mathbf{F}_{\mathsf{F}}^{-1} = 1$. The all-NS superconformal block $\mathbf{F}_a$ can then be computed as

$$\begin{aligned}\mathbf{F}_a(h_1,h_2,h_3,c) = (\mathbf{F}_{\mathsf{F}})^{-1} \star & \sum_{\substack{n_1,n_2,n_3\in\frac{1}{2}\mathbb{Z} \\ (-1)^{2(n_1+n_2+n_3)}=(-1)^a}} q_1^{2n_1^2} q_2^{2n_2^2} q_3^{2n_3^2} \eta_1^{2n_1+p_1} \eta_2^{2n_2+p_2} \eta_3^{2n_3+p_3} \\ & \times (-1)^{(2n_1+p_1)(2n_2+p_2)+(2n_1+p_1)(2n_3+p_3)+(2n_2+p_2)(2n_3+p_3)} \\ & \times [B_a(h_1,h_2,h_3;n_1,n_2,n_3)]^2 F(h^{(1)}_{1,n_1}, h^{(1)}_{2,n_2}, h^{(1)}_{3,n_3}, c^{(1)}) F(h^{(2)}_{1,n_1}, h^{(2)}_{2,n_2}, h^{(2)}_{3,n_3}, c^{(2)}).\end{aligned} \tag{7.27}$$

The NS branching coefficient

The remaining problem is to compute the branching coefficient $B_a(h_1,h_2,h_3;n_1,n_2,n_3)$. This has been worked out in [**1312.4520**, **1406.3008**] (they called it the blow up factor). To compute this factor, let us define the normalization of v_n by the following.

We first introduce a free field representation of the $\mathsf{SCA}_{1/2}$ algebra. The generators of the free field algebra $\mathcal{A}_{1/2}$ involve free boson operators $\{c_n | n \neq 0\}$, free fermion operators $\left\{\eta_r | r \in \mathbb{Z}+\frac{1}{2}\right\}$ (called ψ_r in [**1608.02568**]) and the central elements $\hat{P}$. They satisfy the algebraic relations

$$[c_n, c_m] = n\delta_{m+n,0}, \;\; \{\eta_r, \eta_s\} = \delta_{r+s,0}. \tag{7.28}$$

We denote by $\mathcal{V}(P)$ the Fock representation of the free-field algebra on which the central element $\hat{P}$ acts as P. It is generated by acting c_{-n}, η_{-r} on the lowest weight vector $\phi(P)$, with $\hat{P}\phi(P) = P\phi(P)$.

The free field representation of $\mathsf{SCA}_{1/2}$ is obtained by

$$\begin{aligned} L_n &= \frac{1}{2}\sum_{k\neq 0,n} c_k c_{n-k} + \frac{1}{2}\sum_r r\eta_{n-r}\eta_r + \frac{i}{2}(Qn - 2\hat{P})c_n, \quad n \neq 0, \\ L_0 &= \sum_{k>0} c_{-k}c_k + \sum_{r>0} r\eta_{-r}\eta_r + \frac{1}{2}\left(\frac{Q^2}{4} - \hat{P}^2\right), \\ G_r &= \sum_{n\neq 0} c_n \eta_{r-n} + i(Qr - \hat{P})\eta_r. \end{aligned} \tag{7.29}$$

The Fock representation $\mathcal{V}(P)$ is therefore also a representation of $\mathsf{SCA}_{1/2}$. For $P \notin \frac{1}{2}(b\mathbb{Z}+b^{-1}\mathbb{Z})$, this representation is irreducible as a $\mathsf{SCA}_{1/2}$ representation and is isomorphic to the highest weight representation $\mathcal{V}_{\mathsf{SCA}_{1/2}}(h)$ with $h = \frac{1}{8}Q^2 - \frac{1}{2}P^2$.

Consider the tensor-product representation $\mathcal{M}(P) := \mathcal{V}_{\mathsf{F}_{1/2}} \otimes \mathcal{V}(P)$, where $\mathcal{V}_{\mathsf{F}_{1/2}}$ is the Fock representation of the auxiliary free fermion. Since $\mathcal{M}(P)$ is a representation of $\mathsf{F}_{1/2} \oplus \mathsf{SCA}_{1/2}$ and two copy of virasoro algebra $\mathsf{Vir} \oplus \mathsf{Vir}$ is a subalgebra of $\mathsf{F}_{1/2} \oplus \mathsf{SCA}_{1/2}$, it is a representation of $\mathsf{Vir} \oplus \mathsf{Vir}$.

For $P \notin \frac{1}{2}(b\mathbb{Z}+b^{-1}\mathbb{Z})$, $\mathcal{M}(P)$ is isomorphic to $\mathcal{M}_{NS}(h)$ with $h=\frac{1}{8}Q^2-\frac{1}{2}P^2$. Under this identification, $\mathcal{M}(P)$ admits two natural bases. One is the Fock basis inherited from $\mathcal{M}(P)$, obtained by acting the free field generators $c_{-n}, \eta_{-r}, \psi_{-s}$ on the lowest weight vector $\mathbf{1}_{\mathsf{F}}\otimes\phi(P)$. Another basis is the PBW basis inherited from $\mathcal{M}_{NS}(h)$, obtained by acting the super-virasoro generators $L_{-n}, G_{-r}, \psi_{-s}$ on $\mathbf{1}_{\mathsf{F}}\otimes\phi(P)$. One can thus consider the base transformation between Fock basis and PBW basis, which a prior would be complicated.

For our application, we will always endow $\mathcal{M}(P)$ with the BPZ pairing, under which $L_n^\dagger = L_{-n}, G_r^\dagger = G_{-r}$. Under this pairing, one can speak of the BPZ conjugation of free field generators c_n, η_r, which generically does not admit a simple formula in terms of the original free field generators. Due to this reason, it is not straightforward to compute the norm of the state in the Fock basis. The efficient way to compute the norm is to do the base transformation to the PBW basis and then compute the norm there.

Yutai: The derivation in the NS sector yet to be reviewed. The point seems to be that the BPZ pairing is better defined as the pairing between $\mathcal{M}(P)$ and $\mathcal{M}(-P)$. The BPZ conjugate then maps the super-Virasoro generator in one representation to another (isomorphic) representation. It is different from the usual representation theory setup where the pairing/inner product is defined on the same space.

The $\mathsf{Vir}\oplus\mathsf{Vir}$ primaries for $n>0$ in $\mathcal{M}(P)$ admits a simple formula in the Fock basis

$$w_n(P) = \chi_{-\frac{4n-1}{2}}(P)\chi_{-\frac{4n-3}{2}}(P)\cdots\chi_{-\frac{1}{2}}(P)\phi(P), \quad \chi_r = \psi_r - i\eta_r. \tag{7.30}$$

For each free boson mode c_n and free fermion mode η, one can directly solve them in terms of G_{-r}, L_{-n} and $\hat{P}$. We define the Virasoro primary w_n in $\mathcal{M}_{\mathrm{NS}}(h)$ to be the state after replacing c,η by L, G modes, $\phi(P)$ by ϕ, and $\hat{P}$ takes the positive root of $\frac{1}{4}Q^2-2h$. If h is greater than $Q^2/8$, we take $P=ip$ such that $p>0$. The two states $w_n(P)$ and w_n will have the same norm.

Under the BPZ inner product where $L_n^\dagger = L_{-n}, G_r^\dagger = G_{-r}$, we adopt the following conjugation rules for $c, \eta, \hat{P}$:

$$c_n^\dagger = c_{-n}, \quad \eta_r^\dagger = \eta_{-r}, \quad \hat{P}^\dagger = -\hat{P} \tag{7.31}$$

and hence the BPZ conjugate of $\chi_r(P)$ is $-\chi_{-r}(-P) = -\psi_{-r}+i\eta_{-r}(-P)$. The remaining step of computing $\|w_n\|^2$ is then determining the commutator $\{\chi_r(-P), \chi_s(P)\}$. This is worked out in [**1312.4520**] by analyzing the pole structure of this quantity in P, and it turns out that

$$\|w_n(P)\|^2 = (-1)^{2n}2^{2n}\frac{\ell(2P,4n)}{\ell(Q+2P,4n)} \tag{7.32}$$

where the ℓ function is defined by the following. For positive integer m,

$$\ell(x,m) = \begin{cases} \prod_{\substack{r,s\geq 0, r+s<m \\ r+s\equiv 0(2)}} (x+rb+sb^{-1}), & m>0 \text{ even} \\ 2^{1/8}\prod_{\substack{r,s\geq 0, r+s<m \\ r+s\equiv 1(2)}} (x+rb+sb^{-1}), & m>0 \text{ odd} \end{cases} \tag{7.33}$$

For zero or negative integer m,

$$\ell(x,0)=1, \quad \ell(x,m)=\begin{cases}(-1)^{-m/2}\ell(Q-x,-m), & m<0 \text{ even}\\ \ell(Q-x,-m), & m<0 \text{ odd}\end{cases} \tag{7.34}$$

This formula has been checked against direct computation of the state w_n up to level $n=\frac{3}{2}$.

We normalize the Virasoro primary v_n with positive n to be

$$v_n = 2^{-2n}\ell(Q+2P,4n)w_n, \tag{7.35}$$

where, again, P is taken to be the positive for small h and purely imaginary with positive imaginary part for large h. v_0 is defined to be ϕ, and v_n with negative n is defined from the reflection formula

$$v_n(P)=v_{-n}(-P). \tag{7.36}$$

The norm of v_n is given by

$$\|v_n\|^2 = (-1)^{2n}2^{-2n}\ell(2P,4n)\ell(Q+2P,4n). \tag{7.37}$$

The 3-point function $\hat{\rho}_a(v_{1,n_1},v_{2,n_2},v_{3,n_3})$ is also computed in [**1312.4520**]. Normalizing

$$\hat{\rho}_0(\phi_1,\phi_2,\phi_3)=1, \quad \hat{\rho}_1(\phi_1,G_{-1/2}\phi_2,\phi_3)=1. \tag{7.38}$$

In our convention, their result reads

$$\begin{aligned}\hat{\rho}_a(v_{1,n_1},v_{2,n_2},v_{3,n_3}) &= (-1)^{2n_3}(-1)^{n_1+n_2+n_3}i^a 2^{-(n_1+n_2+n_3)}\\ &\times \ell\left(\tfrac{Q}{2}+P_1+P_2+P_3,2(n_1+n_2+n_3)\right)\ell\left(\tfrac{Q}{2}-P_1+P_2+P_3,2(-n_1+n_2+n_3)\right)\\ &\times \ell\left(\tfrac{Q}{2}+P_1+P_2-P_3,2(n_1+n_2-n_3)\right)\ell\left(\tfrac{Q}{2}-P_1+P_2-P_3,2(-n_1+n_2-n_3)\right).\end{aligned} \tag{7.39}$$

For $(-1)^{2(n_1+n_2+n_3)}\neq(-1)^a$, the 3-point function is 0 by definition. Again, we should emphasize that this result only works for positive n_i. For negative n_i, one should use the reflection relation $v_n(P)=v_{-n}(-P)$ to convert it to a positive value. The values of P_i is the same in the definition of v_n. This formula has been checked directly against low-level ($n=0,1,\frac{1}{2}$) 3-point functions.

Putting everything together, we arrive at the final form for the NS branching coefficient:

$$\begin{aligned}B_a(h_1,h_2,h_3;n_1,n_2,n_3) &= \frac{(-1)^{2n_3}(-i)^a}{\prod_{i=1}^3\sqrt{\ell(2P_i,4n_i)\ell(Q+2P_i,4n_i)}}\\ &\times \ell\left(\tfrac{Q}{2}+P_1+P_2+P_3,2(n_1+n_2+n_3)\right)\ell\left(\tfrac{Q}{2}-P_1+P_2+P_3,2(-n_1+n_2+n_3)\right)\\ &\times \ell\left(\tfrac{Q}{2}+P_1+P_2-P_3,2(n_1+n_2-n_3)\right)\ell\left(\tfrac{Q}{2}-P_1+P_2-P_3,2(-n_1+n_2-n_3)\right).\end{aligned} \tag{7.40}$$

A Recursion Algorithm For Branching Coefficient

In this subsection, we explore a recursive algorithm for constructing the double Virasoro primary in the PBW basis. The idea is to study the state $L_1 |w_n\rangle$. In the following, we will prove that $L_1 w_n$ is the double Virasoro descendant of w_{n-1}. Namely, it admits an expansion as

$$L_1 |w_n\rangle = \sum_{|A|+|B|=4n-3} C_{AB} L^{(1)}_{-A} L^{(2)}_{-B} |w_{n-1}\rangle \,. \tag{7.41}$$

Meanwhile, following from the free field representation, we can show that the following linear equations uniquely determines $L_1 |w_n\rangle$ up to a constant:

$$J_{-\mu} L_1 |w_n\rangle = 2^{\ell(\mu)} |w_{n-1}\rangle \,, \quad c_m L_1 |w_n\rangle = 0. \tag{7.42}$$

It allows us to pin down the coefficient C_{AB}.

To prove the relations above, introduce

$$\chi(z) = \frac{1}{\sqrt{2}} \left(\psi(z) - i\eta(z)\right), \; \chi(z)^* = \frac{1}{\sqrt{2}} \left(\psi(z) + i\eta(z)\right). \tag{7.43}$$

The $U(1)$ current for the complex fermion is given by

$$J(z) = \frac{1}{2} : \chi(z)\chi^* z : \tag{7.44}$$

with mode expansion

$$J(z) = \sum_n J_n z^{-n-1}, \quad J_n = \frac{1}{2} \sum_{r\in\mathbb{Z}+\frac{1}{2}} : \chi_{n-r}\chi^*_r : . \tag{7.45}$$

It satisfy

$$[J_m, J_n] = m\delta_{m+n,0}, \quad [J_m, \chi_r] = \chi_r, \quad [J_m, \chi^*_r] = -\chi^*_r. \tag{7.46}$$

As w_n is constructed by filling the fermi sea, we have

$$c_m |w_n\rangle = 0, \tag{7.47}$$

Using the free field representation, we can show that

$$J_{-\mu} L_1 |w_n\rangle = . \tag{7.48}$$

where J_n is the $U(1)$ current generators

$$J_n =: \chi_r \chi^*_{n-r} : \tag{7.49}$$

The coefficient C_{AB} can be extracted from the matrix element

$$\left\langle L^{(1)}_{-A'}L^{(2)}_{-B'}w_{n-1}\middle|L_1 w_n\right\rangle = \left\langle L_{-1}L^{(1)}_{-A'}L^{(2)}_{-B'}w_{n-1}\middle|w_n\right\rangle \tag{7.50}$$

We compute this matrix element by expressing both side as PBW basis acting on w_{n-1}

$$\left\langle L^{(1)}_{-A'}L^{(2)}_{-B'}w_{n-1}\middle|L_1 w_n\right\rangle = \sum_{|A|+|B|=4n-3} C_{AB}||w_{n-1}||^2\mathbb{G}_{AA'}\mathbb{G}_{BB'} \tag{7.51}$$

Combine the fermion ψ, η into one complex fermion χ, with

$$\chi_r = \frac{1}{\sqrt{2}}(\psi_r - i\eta_r), \quad \chi^*_r = \frac{1}{\sqrt{2}}(\psi_r + i\eta_r). \tag{7.52}$$

They will satisfy the commutation relation

$$\{\chi_r, \chi^*_s\} = \delta_{r+s,0}. \tag{7.53}$$

In the free field representation, the double virasoro primary can be represented by

$$w_n(P) = \chi_{-\frac{4n-1}{2}}(P)\chi_{-\frac{4n-3}{2}}(P)\cdots\chi_{-\frac{1}{2}}(P)\phi(P). \tag{7.54}$$

Checks against c-recursion

As a check of the formalism developed in this section, we compute the all-NS block $\mathbf{F}_a(h_1, h_2, h_3, c)$ as a q-expansion independently using the c-recursion relation and the double Virasoro decomposition. For all the vacuum blocks appearing in $\mathbf{F}, \mathbf{F}_{\mathsf{F}}$ and F, we use the Schottky frame so that the Weyl anomaly factor on both sides agrees. We have tested numerically that, up to total level 4, the all-NS theta channel block $\mathbf{F}_a$ computed from the two methods agrees to machine precision for all 8 independent choices of $\{\eta_i\}$ and a.

8 Ramond blocks from double Virasoro

We now generalize the discussion above to Ramond blocks. Denoting the fermion ground states by u^{a} with $\mathsf{a} = 0, 1$, normalizing

$$\langle u^0|u^0\rangle = 1, \quad \langle u^1|u^1\rangle = -1, \quad \psi_0 u^{\mathsf{a}} = \frac{1}{\sqrt{2}}u^{1-\mathsf{a}}, \tag{8.1}$$

the ramond $\mathsf{SCA}_0 \otimes \mathsf{F}_0$ block is defined as

$$
\begin{aligned}
\widehat{\mathbb{F}}_f^{(\eta,\eta')}(h_1,\beta_2,\beta_3,c) = & \sum_{\substack{|A|=|A'|,|B|=|B'|,|C|=|C'| \\ (-1)^{A+B+C+|\alpha|+|\gamma|}=(-1)^f \\ (-1)^{A'+B'+C'+|\alpha'|+|\gamma'|}=(-1)^f \\ (-1)^{\mathsf{A}+\mathsf{B}+\mathsf{C}+\mathsf{b}+\mathsf{c}}=1}} q_1^{|A|}\eta_1^{A+p_1} q_2^{|B|}\eta_2^{B+|\alpha|} q_3^{|C|}\eta_3^{C+|\gamma|} \mathbf{G}_{h_1}^{AA'} \mathbb{G}_{\beta_2}^{B\alpha,B'\alpha'} \mathbb{G}_{\beta_3}^{C\gamma,C'\gamma'} \\
& \times (-1)^{\mathrm{sgn}(A)\mathrm{sgn}(B)+\mathrm{sgn}(A)\mathrm{sgn}(C)+\mathrm{sgn}(B)\mathrm{sgn}(C)+A+\mathsf{A}} q_1^{|\mathsf{A}|}\eta_1^{\mathsf{A}} q_2^{|\mathsf{B}|}\eta_2^{\mathsf{B}+\mathsf{b}} q_3^{|\mathsf{C}|}\eta_3^{\mathsf{C}+\mathsf{c}} \\
& \times \hat{\rho}_f^{(\eta)}(\Psi_{-\mathsf{A}}\mathbf{L}_{-A}\phi_1, \boldsymbol{\Psi}_{-\mathsf{B}}u^{\mathsf{b}} \otimes \mathbb{L}_{-B}w_2^{\alpha}, \boldsymbol{\Psi}_{-\mathsf{C}}u^{\mathsf{c}} \otimes \mathbb{L}_{-C}w_3^{\gamma}) \\
& \times \hat{\rho}_f^{(\eta')}(\Psi_{-\mathsf{A}}\mathbf{L}_{-A'}\phi_1, \boldsymbol{\Psi}_{-\mathsf{B}}u^{\mathsf{b}} \otimes \mathbb{L}_{-B'}w_2^{\alpha'}, \boldsymbol{\Psi}_{-\mathsf{C}}u^{\mathsf{c}} \otimes \mathbb{L}_{-C'}w_3^{\gamma'})
\end{aligned}
\tag{8.2}
$$

where $\boldsymbol{\Psi}_{-\mathsf{B}}$ denotes a chain of R-sector fermion lowering operators (there are no `mathbb` version of Ψ, sadly). Note that $\boldsymbol{\Psi}_{-\mathsf{A}}$ does not contain the fermion zero mode ψ_0. We have used $\langle u^{\mathsf{a}}|\boldsymbol{\Psi}_{-\mathsf{A}}^{\dagger}\boldsymbol{\Psi}_{-\mathsf{B}}|u^{\mathsf{b}}\rangle = (-1)^{\mathsf{A}+\mathsf{a}}\delta_{\mathsf{AB}}\delta^{\mathsf{ab}}$, as well as $\langle u_1 \otimes x_1|u_2 \otimes x_2\rangle = \langle u_1|u_2\rangle\langle x_1|x_2\rangle$ without the additional operator exchanging sign. The shorthand sgn is defined such that $\mathrm{sgn}(A) = p_1+A+\mathsf{A} \mod 2$ for NS sector and $\mathrm{sgn}(B) = B+\mathsf{B}+\mathsf{b}+|\alpha| \mod 2$ for R sector. We have also used the fact that the fermion correlator is 0 unless $(-1)^{\mathsf{A}+\mathsf{B}+\mathsf{C}+\mathsf{b}+\mathsf{c}} = 1$.

The 3-point function can be factorized as (we treat this formula as the definition of $\hat{\rho}_f^{(\eta)}$)

$$
\begin{aligned}
&\hat{\rho}_f^{(\eta)}(\Psi_{-\mathsf{A}}\mathbf{L}_{-A}\phi_1, \boldsymbol{\Psi}_{-\mathsf{B}}u^{\mathsf{b}} \otimes \mathbb{L}_{-B}w_2^{\alpha}, \boldsymbol{\Psi}_{-\mathsf{C}}u^{\mathsf{c}} \otimes \mathbb{L}_{-C}w_3^{\gamma}) \\
&\quad = (-1)^{A\mathsf{A}+(B+|\alpha|)(\mathsf{C}+\mathsf{c})+p_1(\mathsf{C}+\mathsf{c})}\rho(\Psi_{-\mathsf{A}}\mathbf{1}_{\mathsf{F}}, \boldsymbol{\Psi}_{-\mathsf{B}}u^{\mathsf{b}}, \boldsymbol{\Psi}_{-\mathsf{C}}u^{\mathsf{c}})\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^{\alpha}, \mathbb{L}_{-C}w_3^{\gamma}),
\end{aligned}
\tag{8.3}
$$

where the fermionic 3-point function is normalized such that $\rho(\mathbf{1}_{\mathsf{F}}, u^0, u^0) = 1, \rho(\mathbf{1}_{\mathsf{F}}, u^1, u^1) = i$. Therefore, if we define the fermion Ramond block as

$$
\begin{aligned}
\mathbb{F}_{\mathsf{F}} = \sum_{\mathsf{A}+\mathsf{B}+\mathsf{C}+\mathsf{b}+\mathsf{c}\equiv 0 \mod 2} & (-1)^{\mathsf{A}} q_1^{|\mathsf{A}|}\eta_1^{\mathsf{A}} q_2^{|\mathsf{B}|}\eta_2^{\mathsf{B}+\mathsf{b}} q_3^{|\mathsf{C}|}\eta_3^{\mathsf{C}+\mathsf{c}} (-1)^{\mathsf{A}(\mathsf{B}+\mathsf{b})+\mathsf{A}(\mathsf{C}+\mathsf{c})+(\mathsf{B}+\mathsf{b})(\mathsf{C}+\mathsf{c})} \\
& \times [\rho(\Psi_{-\mathsf{A}}\mathbf{1}_{\mathsf{F}}, \boldsymbol{\Psi}_{-\mathsf{B}}u^{\mathsf{b}}, \boldsymbol{\Psi}_{-\mathsf{C}}u^{\mathsf{c}})]^2,
\end{aligned}
\tag{8.4}
$$

one then have the decomposition

$$
\widehat{\mathbb{F}}_f^{(\eta,\eta')}(h_1,\beta_2,\beta_3,c) = \mathbb{F}_{\mathsf{F}} \star_{\mathrm{R}} \mathbb{F}_f^{(\eta,\eta')}(h_1,\beta_2,\beta_3,c)
$$

where the R-sector convolution acts on monomials of q_i and η_i as

$$
\begin{aligned}
& q_1^{r_1}q_2^{r_2}q_3^{r_3}\eta_1^{\epsilon_1}\eta_2^{\epsilon_2}\eta_3^{\epsilon_3} \star_{\mathrm{R}} q_1^{s_1}q_2^{s_2}q_3^{s_3}\eta_1^{\delta_1}\eta_2^{\delta_2}\eta_3^{\delta_3} \\
&\qquad = (-1)^{\sum_{i<j}(\epsilon_i\delta_j+\delta_i\epsilon_j)} q_1^{r_1+s_1}q_2^{r_2+s_2}q_3^{r_3+s_3}\eta_1^{\epsilon_1+\delta_1}\eta_2^{\epsilon_2+\delta_2}\eta_3^{\epsilon_3+\delta_3}.
\end{aligned}
\tag{8.5}
$$

Using the decomposition (7.15), one can see that every state in $\mathcal{M}(\beta)$ can be written as $L_{-A}^{(1)}L_{-B}^{(2)}v_n^{\alpha}$, with $\alpha = \pm$ and $n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}$. The state v_n^{α} has conformal weight $h_n^{(1)} + h_n^{(2)} - h =$

$2n^2 - \frac{1}{16}$ and fermion parity $(-1)^\alpha$. Therefore,

$$\begin{aligned}
\widehat{\mathbb{F}}_f^{(\eta,\eta')}(h_1,\beta_2,\beta_3,c) = &\sum_{\substack{|A|=|A'|,|B|=|B'|,|C|=|C'|\\ |D|=|D'|,|E|=|E'|,|F|=|F'|\\ n_1\in\frac{1}{2}\mathbb{Z},\ n_2,n_3\in\frac{1}{2}\mathbb{Z}+\frac{1}{4}\\ (-1)^{2n_1+\alpha_2+\alpha_3}=(-1)^f}} q_1^{2n_1^2+|A|+|B|} q_2^{2n_2^2-\frac{1}{8}+|C|+|D|} q_3^{2n_3^2-\frac{1}{8}+|E|+|F|} \eta_1^{2n_1+p_1}\eta_2^{\alpha_2}\eta_3^{\alpha_3} \\
&\times (-1)^{(2n_1+p_1)\alpha_2+(2n_1+p_1)\alpha_3+\alpha_2\alpha_3} \frac{(G_{h^{(1)}_{1,n_1}})^{AA'}(G_{h^{(2)}_{1,n_1}})^{BB'}(G_{h^{(1)}_{2,n_2}})^{CC'}(G_{h^{(2)}_{2,n_2}})^{DD'}(G_{h^{(1)}_{3,n_3}})^{EE'}(G_{h^{(2)}_{3,n_3}})^{FF'}}{\|v^{p_1}_{1,n_1}\|^2\|v^{\alpha_2}_{2,n_2}\|^2\|v^{\alpha_3}_{3,n_3}\|^2} \\
&\times \hat{\rho}_f^{(\eta)}(L^{(1)}_{-A}L^{(2)}_{-B}v^{p_1}_{1,n_1}, L^{(1)}_{-C}L^{(2)}_{-D}v^{\alpha_2}_{2,n_2}, L^{(1)}_{-E}L^{(2)}_{-F}v^{\alpha_3}_{3,n_3})\hat{\rho}_f^{(\eta')}(L^{(1)}_{-A'}L^{(2)}_{-B'}v^{p_1}_{1,n_1}, L^{(1)}_{-C'}L^{(2)}_{-D'}v^{\alpha_2}_{2,n_2}, L^{(1)}_{-E'}L^{(2)}_{-F'}v^{\alpha_3}_{3,n_3}),
\end{aligned} \tag{8.6}$$

where we used the fact that u^{a} has conformal weight $\frac{1}{16}$ and the fact that $\langle v_n^0|v_n^1\rangle = 0$. The factorization relation of the 3-point function reads

$$\begin{aligned}
&\hat{\rho}_f^{(\eta)}(L^{(1)}_{-A}L^{(2)}_{-B}v^{p_1}_{1,n_1}, L^{(1)}_{-C}L^{(2)}_{-D}v^{\alpha_2}_{2,n_2}, L^{(1)}_{-E}L^{(2)}_{-F}v^{\alpha_3}_{3,n_3}) \\
&\quad = \hat{\rho}_f^{(\eta)}(v^{p_1}_{1,n_1}, v^{\alpha_2}_{2,n_2}, v^{\alpha_3}_{3,n_3})\rho(L_{-A}\nu_{h^{(1)}_{1,n_1}}, L_{-C}\nu_{h^{(1)}_{2,n_2}}, L_{-E}\nu_{h^{(1)}_{3,n_3}})\rho(L_{-B}\nu_{h^{(2)}_{1,n_1}}, L_{-D}\nu_{h^{(2)}_{2,n_2}}, L_{-F}\nu_{h^{(2)}_{3,n_3}}),
\end{aligned} \tag{8.7}$$

and therefore,

$$\begin{aligned}
\widehat{\mathbb{F}}_f^{(\eta,\eta')}(h_1,\beta_2,\beta_3,c) = &\sum_{\substack{n_1\in\frac{1}{2}\mathbb{Z}, n_2,n_3\in\frac{1}{2}\mathbb{Z}+\frac{1}{4}\\ (-1)^{2n_1+\alpha_2+\alpha_3}=(-1)^f}} q_1^{2n_1^2} q_2^{2n_2^2-\frac{1}{8}} q_3^{2n_3^2-\frac{1}{8}} \eta_1^{2n_1+p_1}\eta_2^{\alpha_2}\eta_3^{\alpha_3} \\
&\times (-1)^{2n_1+(2n_1+p_1)\alpha_2+(2n_1+p_1)\alpha_3+\alpha_2\alpha_3} \\
&\times \mathbb{B}_f^{(\eta);p_1}(h_1,\beta_2,\beta_3;n_1,n_2,n_3)\mathbb{B}_f^{(\eta');p_1}(h_1,\beta_2,\beta_3;n_1,n_2,n_3) \\
&\times F(h^{(1)}_{1,n_1}, h^{(1)}_{2,n_2}, h^{(1)}_{3,n_3}, c^{(1)})F(h^{(2)}_{1,n_1}, h^{(2)}_{2,n_2}, h^{(2)}_{3,n_3}, c^{(2)}),
\end{aligned} \tag{8.8}$$

where the R-sector branching coefficients are defined as

$$\mathbb{B}_f^{(\eta);p_1}(h_1,\beta_2,\beta_3;n_1,n_2,n_3) = \frac{\hat{\rho}_f^{(\eta)}(v^{p_1}_{1,n_1}, v^{\alpha_2}_{2,n_2}, v^{\alpha_3}_{3,n_3})}{\|v^{p_1}_{1,n_1}\|\|v^{\alpha_2}_{2,n_2}\|\|v^{\alpha_3}_{3,n_3}\|}. \tag{8.9}$$

where $v^{p_1}_{1,n_1}$ has parity $2n_1 + p_1$.

Ramond branching coefficients

Unlike the NS branching coefficient $\mathbb{B}_a$, the Ramond branching coefficient $\mathbb{B}_f^{(\eta);p_1}$ is not computed in the literature. There are conjectures in [**1312.4520**] about its structure, but the numerical prefactor remains unknown. In the rest of this section, we develop a recursion relation between $\mathbb{B}_f^{(\eta);p_1}(n_1,n_2,n_3,\alpha_2,\alpha_3)$'s, which can be used to solve them numerically.

Let's first define the Virasoro primaries v_n^α unambiguously. Similar to the NS algebra, one can also introduce free bosons c_n and free fermions η_r with $n, r \in \mathbb{Z}$ to realize the SCA_0 algebra. The SCA generators are given by the same formula (**??**). Denoting $\eta_r(P)$ as the operator η_r expressed using G_r, L_n, P. Using $\chi_r(P) = \psi_r - i\eta_r(P)$, we define v_n^α with $n > 0$ as

$$\begin{aligned} v_n^{2n+\frac{1}{2} \bmod 2}(P) &= \chi_0(P)\chi_{-1}(P) \cdots \chi_{-(2n-\frac{1}{2})}(P)(u^0 \otimes w_\beta^+), \\ v_n^{2n+\frac{1}{2} \bmod 2}(P) &= \chi_0(P)\chi_{-1}(P) \cdots \chi_{-(2n-\frac{1}{2})}(P)\chi_0(-P)(u^0 \otimes w_\beta^+), \end{aligned} \tag{8.10}$$

where $\chi_0(-P) = \psi_0 + i\eta_0(P)$. The states with $n < 0$ are defined using

$$(v_n^\alpha(P))_+ = (v_{-n}^\alpha(-P))_+, \quad (v_n^\alpha(P))_- = -(v_{-n}^\alpha(-P))_-, \tag{8.11}$$

where the subscripts $\pm$ denote the components ending in w_β^+ and w_β^-, respectively. The BPZ norms of the Virasoro primaries can be computed by

$$\begin{aligned} \left\| v_n^0(P) \right\|^2 &= 2^{2\lfloor M/2 \rfloor + 1} \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}, \\ \left\| v_n^1(P) \right\|^2 &= -2^{2\lceil M/2 \rceil} \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}, \end{aligned} \tag{8.12}$$

where $M = 2n - \frac{1}{2}$.

The key observation that leads to the recursion relation is the following:

Proposition 1. *For Ramond* $\mathsf{Vir} \oplus \mathsf{Vir}$ *primaries with* $n \geq \frac{3}{4}$, $L_1 v_n^\alpha$ *only contains level-*$(4n-3)$ $\mathsf{Vir} \oplus \mathsf{Vir}$ *descendants of* v_{n-1}^α, *and* $L_{-1} v_n^\alpha$ *only contains level-1* $\mathsf{Vir} \oplus \mathsf{Vir}$ *descendants of* v_n^α *and level-*$(4n-1)$ $\mathsf{Vir} \oplus \mathsf{Vir}$ *descendants of* v_{n-1}^α. *Namely,*

$$\begin{aligned} L_1 v_n^\alpha &= \sum_{|A|+|B|=4n-3} \mathbb{V}_{1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^\alpha, \\ L_{-1} v_n^\alpha &= \mathbb{V}_{n,\alpha}^{(1)} L_{-1}^{(1)} v_n^\alpha + \mathbb{V}_{n,\alpha}^{(2)} L_{-1}^{(2)} v_n^\alpha + \sum_{|A|+|B|=4n-1} \mathbb{V}_{-1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^\alpha, \end{aligned} \tag{8.13}$$

for some $\mathbb{V}_{\pm 1,n,\alpha}^{AB}$ *and* $\mathbb{V}_{n,\alpha}^{(1,2)}$.

This has been checked directly on $n = \frac{3}{4}, \frac{5}{4}, \frac{7}{4}$ and $\frac{9}{4}$ by directly acting $L_{\pm 1}$ on the states and project onto the subspace on the RHS. For NS sector, one have a similar relation:

Proposition 2. *For NS* $\mathsf{Vir} \oplus \mathsf{Vir}$ *primaries with* $n \geq 1$, $L_1 v_n$ *only contains level-*$(4n-3)$ $\mathsf{Vir} \oplus \mathsf{Vir}$ *descendants of* v_{n-1}, *and* $L_{-1} v_n$ *only contains level-1* $\mathsf{Vir} \oplus \mathsf{Vir}$ *descendants of* v_n *and level-*$(4n-1)$ $\mathsf{Vir} \oplus \mathsf{Vir}$ *descendants of* v_{n-1}. *Namely,*

$$\begin{aligned} L_1 v_n &= \sum_{|A|+|B|=4n-3} \mathbf{V}_{1,n}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}, \\ L_{-1} v_n &= \mathbf{V}_n^{(1)} L_{-1}^{(1)} v_n + \mathbf{V}_n^{(2)} L_{-1}^{(2)} v_n + \sum_{|A|+|B|=4n-1} \mathbf{V}_{-1,n}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}. \end{aligned} \tag{8.14}$$

We have checked this for $n = 1, \frac{3}{2}, 2, \frac{5}{2}$. Note that these coefficients $\mathbb{V}$ and $\mathbf{V}$'s will depend on P and c in general. These coefficients can be solved efficiently. For a generic point of (P, c) and on a laptop, solving all coefficients for R-sector $n = \frac{9}{4}$ takes around 3 seconds, and NS-sector $n = \frac{5}{2}$ takes around 8 seconds.

After solving the coefficients $\mathbb{V}$ and $\mathbf{V}$'s, one can use the L_1 conformal Ward identities:

$$\begin{aligned}
\hat{\rho}_f^{(\eta)}(L_1 v_{1,n_1}^{p_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) &= \hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, L_{-1} v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) + \hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, v_{2,n_2}^{\alpha_2}, L_{-1} v_{3,n_3}^{\alpha_3}), \\
\hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, L_1 v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) &= \hat{\rho}_f^{(\eta)}(L_{-1} v_{1,n_1}^{p_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) - \hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, L_{-1} v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) \\
&\quad - 2\hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, L_0 v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) - \hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, v_{2,n_2}^{\alpha_2}, L_1 v_{3,n_3}^{\alpha_3}).
\end{aligned} \tag{8.15}$$

Plugging in the decomposition above, the first equation then reads

$$\begin{aligned}
&\sum_{|A|+|B|=4n_1-3} \mathbf{V}_{1,n_1}^{AB} \rho(L_{-A}\nu_{h_{1,n_1-1}^{(1)}}, \nu_{h_{2,n_2}^{(1)}}, \nu_{h_{3,n_3}^{(1)}}) \rho(L_{-B}\nu_{h_{1,n_1-1}^{(2)}}, \nu_{h_{2,n_2}^{(2)}}, \nu_{h_{3,n_3}^{(2)}}) \hat{\rho}_f^{(\eta)}(v_{1,n_1-1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) \\
&\quad = \left[\sum_{i=1}^{2} (\mathbb{V}_{n_2,\alpha_2}^{(i)} - \mathbb{V}_{n_3,\alpha_3}^{(i)})(h_{1,n_1}^{(i)} - h_{2,n_2}^{(i)} - h_{3,n_3}^{(i)})\right] \hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) \\
&\quad + \sum_{|C|+|D|=4n_2-1} \mathbb{V}_{-1,n_2,\alpha_2}^{CD} \rho(\nu_{h_{1,n_1}^{(1)}}, L_{-C}\nu_{h_{2,n_2-1}^{(1)}}, \nu_{h_{3,n_3}^{(1)}}) \rho(\nu_{h_{1,n_1}^{(2)}}, L_{-D}\nu_{h_{2,n_2-1}^{(2)}}, \nu_{h_{3,n_3}^{(2)}}) \hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, v_{2,n_2-1}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) \\
&\quad + \sum_{|E|+|F|=4n_3-1} \mathbb{V}_{-1,n_3,\alpha_3}^{EF} \rho(\nu_{h_{1,n_1}^{(1)}}, \nu_{h_{2,n_2}^{(1)}}, L_{-E}\nu_{h_{3,n_3-1}^{(1)}}) \rho(\nu_{h_{1,n_1}^{(2)}}, \nu_{h_{2,n_2}^{(2)}}, L_{-F}\nu_{h_{3,n_3-1}^{(2)}}) \hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3-1}^{\alpha_3}),
\end{aligned} \tag{8.16}$$

where we have used the fact that $\rho(\nu_1, L_{-1}\nu_2, \nu_3) = h_1 - h_2 - h_3$, $\rho(\nu_1, \nu_2, L_{-1}\nu_3) = -h_1 + h_2 + h_3$ and the factorization of 3-point functions. Therefore,

$$\begin{aligned}
&\left[\sum_{i=1}^{2} (\mathbb{V}_{n_2,\alpha_2}^{(i)} - \mathbb{V}_{n_3,\alpha_3}^{(i)})(h_{2,n_2}^{(i)} + h_{3,n_3}^{(i)} - h_{1,n_1}^{(i)})\right] \mathbb{B}_f^{(\eta);p_1}(n_1, n_2, n_3, \alpha_2, \alpha_3) = \\
&- \sum_{|A|+|B|=4n_1-3} \mathbf{V}_{1,n_1}^{AB} \rho(L_{-A}\nu_{h_{1,n_1-1}^{(1)}}, \nu_{h_{2,n_2}^{(1)}}, \nu_{h_{3,n_3}^{(1)}}) \rho(L_{-B}\nu_{h_{1,n_1-1}^{(2)}}, \nu_{h_{2,n_2}^{(2)}}, \nu_{h_{3,n_3}^{(2)}}) \mathbb{B}_f^{(\eta);p_1}(n_1 - 1, n_2, n_3, \alpha_2, \alpha_3) \frac{\|v_{n_1-1}\|}{\|v_{n_1}\|} \\
&+ \sum_{|C|+|D|=4n_2-1} \mathbb{V}_{-1,n_2,\alpha_2}^{CD} \rho(\nu_{h_{1,n_1}^{(1)}}, L_{-C}\nu_{h_{2,n_2-1}^{(1)}}, \nu_{h_{3,n_3}^{(1)}}) \rho(\nu_{h_{1,n_1}^{(2)}}, L_{-D}\nu_{h_{2,n_2-1}^{(2)}}, \nu_{h_{3,n_3}^{(2)}}) \mathbb{B}_f^{(\eta);p_1}(n_1, n_2 - 1, n_3, \alpha_2, \alpha_3) \frac{\|v_{n_2-1}^{\alpha_2}\|}{\|v_{n_2}^{\alpha_2}\|} \\
&+ \sum_{|E|+|F|=4n_3-1} \mathbb{V}_{-1,n_3,\alpha_3}^{EF} \rho(\nu_{h_{1,n_1}^{(1)}}, \nu_{h_{2,n_2}^{(1)}}, L_{-E}\nu_{h_{3,n_3-1}^{(1)}}) \rho(\nu_{h_{1,n_1}^{(2)}}, \nu_{h_{2,n_2}^{(2)}}, L_{-F}\nu_{h_{3,n_3-1}^{(2)}}) \mathbb{B}_f^{(\eta);p_1}(n_1, n_2, n_3 - 1, \alpha_2, \alpha_3) \frac{\|v_{n_3-1}^{\alpha_3}\|}{\|v_{n_3}^{\alpha_3}\|},
\end{aligned} \tag{8.17}$$

and the ratios can be computed by

$$\frac{\|v_{n_1-1}\|}{\|v_{n_1}\|} = 2\sqrt{\frac{\ell\,(2P_1, 4n_1 - 4)\,\ell\,(Q + 2P_1, 4n_1 - 4)}{\ell\,(2P_1, 4n_1)\,\ell\,(Q + 2P_1, 4n_1)}}, \quad \frac{\|v_{n_2-1}^{a_2}\|}{\|v_{n_2}^{a_2}\|} = \frac{1}{2}\sqrt{\frac{\ell\,(2P_2, 4n_2 - 4)\,\ell\,(Q + 2P_2, 4n_2)}{\ell\,(Q + 2P_2, 4n_2 - 4)\,\ell\,(2P_2, 4n_2)}}. \tag{8.18}$$

Using this recursion relation, one can solve for the Ramond branching coefficient $\mathbb{B}_f^{(\eta);p_1}(n_1, n_2, n_3, \alpha_2, \alpha_3)$ recursively. One should note that this form only works for $n_1 \geq 1$ and $n_2, n_3 \geq \frac{3}{4}$. At the boundary values, L_1 annihilates $v_{0,\pm\frac{1}{2}}$ and $v^\alpha_{\pm\frac{1}{4}}$. For L_{-1}, one can use

$$\begin{aligned}
L_{-1}v_0 &= L_{-1}^{(1)}v_0 + L_{-1}^{(2)}v_0 \\
L_{-1}v_{1/2} &= -\frac{(2P+b)\left(-2Pb+b^2+1\right)}{2P\left(2Pb+b^2-1\right)}L_{-1}^{(1)}v_{1/2} + \frac{(2Pb+1)\left(-2Pb+b^2+1\right)}{2Pb\left(-2Pb+b^2-1\right)}L_{-1}^{(2)}v_{1/2} \\
&\quad + \frac{b\left(2Pb+b^2+1\right)}{2P\left(-2Pb+b^2-1\right)}L_{-1}^{(1)}v_{-1/2} - \frac{2Pb+b^2+1}{2Pb\left(2Pb+b^2-1\right)}L_{-1}^{(2)}v_{-1/2}, \\
L_{-1}v_{-1/2} &= -\frac{b\left(-2Pb+b^2+1\right)}{2P\left(2Pb+b^2-1\right)}L_{-1}^{(1)}v_{1/2} + \frac{-2Pb+b^2+1}{2Pb\left(-2Pb+b^2-1\right)}L_{-1}^{(2)}v_{1/2} \\
&\quad + \frac{(-2P+b)\left(2Pb+b^2+1\right)}{2P\left(-2Pb+b^2-1\right)}L_{-1}^{(1)}v_{-1/2} + \frac{(2Pb-1)\left(2Pb+b^2+1\right)}{2Pb\left(2Pb+b^2-1\right)}L_{-1}^{(2)}v_{-1/2}. \\
L_{-1}v^\alpha_{1/4} &= -\frac{-4Pb+b^2+2}{2(2Pb-1)}L_{-1}^{(1)}v^\alpha_{1/4} + \frac{-4Pb+2b^2+1}{2b(-2P+b)}L_{-1}^{(2)}v^\alpha_{1/4} \\
&\quad - \frac{\left(-2Pb+b^2+2\right)\left(-2Pb+2b^2+1\right)}{2a+2b(-2P+b)(2Pb-1)}v^\alpha_{-3/4} \\
L_{-1}v^\alpha_{-1/4} &= \frac{4Pb+b^2+2}{2(2Pb+1)}L_{-1}^{(1)}v^\alpha_{-1/4} + \frac{4Pb+2b^2+1}{2b(2P+b)}L_{-1}^{(2)}v^\alpha_{-1/4} \\
&\quad + \frac{\left(2Pb+b^2+2\right)\left(2Pb+2b^2+1\right)}{2^{\alpha+2}b(2P+b)(2Pb+1)}v^\alpha_{3/4}.
\end{aligned} \tag{8.19}$$

Note that the L_{-1} action on $v_{1/4}$ involves $v_{-3/4}$. One can use the similar reduction relation for $L_{\pm1}$ acting on negative n to deal with it. It will turns out that $L_{-1}v_{-3/4}$ only contains level-1 descendant of $v_{-3/4}$ and level-2 descendants of $v_{1/4}$, while $L_1v^\alpha_{-3/4}$ is proportional to $v_{1/4}$. In the end, one only needs to compute a few low-level branching coefficients directly for $n_{\text{NS}} = 0, \pm\frac{1}{2}$ and $n_{\text{R}} = \pm\frac{1}{4}, \pm\frac{3}{4}$, and every other branching coefficient can be obtained using the recursion relation. Using this method, branching coefficients $\mathbb{B}_f^{(\eta')}(2, \frac{7}{4}, \frac{5}{4}, \alpha_2, \alpha_3)$ with all 8 sign configurations takes around 1 second in total on a laptop. The results are also checked against low-level direct calculations for NS $|n| \leq \frac{3}{2}$ and Ramond $|n| \leq \frac{5}{4}$.

One thing worth noticing is that the recursion relation does not depend on the choice of NS primary parity p_1, because all the Virasoro generators is parity even. However, the boundary value of fusion coefficients do depend on the parity, as parity will be present in the conformal ward identity.

Using the branching coefficients, one can compute the full $\mathsf{SCA}_0 \otimes \mathsf{F}_0$ block using (??). We have computed this quantity in two different ways: First by the double Virasoro decomposition, and then by directly computing the blocks $\mathbb{F}_{\mathsf{F}}$ and $\mathbb{F}_f^{(\eta,\eta')}$ by summing over descendants. We have checked that the two blocks agrees up to order q^6. Computing the block $\hat{\mathbb{F}}_0^{(+,+)}$ as a q-

expansion takes 83 seconds on a laptop, in which 75 seconds are used to compute the branching coefficients.

A Fusion polynomials and factorization formulae

We collect the result for all useful fusion polynomials and factorization formulae in this appendix. They are taken directly from the menu of a fusion restaurant in Cambridge, MA. All the formulas here are tested against direct low-level computations and checked to be consistent with the Mobius tranformation that permutes the three slots.

A.1 Fusion polynomials on a (NS, NS, NS) sphere

The All-NS fusion polynomials are defined as

$$\begin{aligned}
\mathbf{P}^0_{rs}(h_i,h_j;c) &= \prod_{\substack{p=1-r,1-r+2,\dots,r-1\\ q=1-s,1-s+2,\dots,s-1\\ p+q-(r+s)\in 4\mathbb{Z}+2}} \frac{\lambda_i-\lambda_j+pb+qb^{-1}}{2\sqrt{2}}\frac{\lambda_i+\lambda_j+pb+qb^{-1}}{2\sqrt{2}},\\
\mathbf{P}^1_{rs}(h_i,h_j;c) &= \prod_{\substack{p=1-r,1-r+2,\dots,r-1\\ q=1-s,1-s+2,\dots,s-1\\ p+q-(r+s)\in 4\mathbb{Z}}} \frac{\lambda_i-\lambda_j+pb+qb^{-1}}{2\sqrt{2}}\frac{\lambda_i+\lambda_j+pb+qb^{-1}}{2\sqrt{2}}.
\end{aligned} \tag{A.1}$$

They satisfy the reflection relation

$$\begin{aligned}
\mathbf{P}^0_{rs}(h_i,h_j;c_{rs}) &= (-1)^{\lceil rs/2\rceil}\mathbf{P}^0_{rs}(h_j,h_i;c_{rs}),\\
\mathbf{P}^1_{rs}(h_i,h_j;c_{rs}) &= (-1)^{\lfloor rs/2\rfloor}\mathbf{P}^1_{rs}(h_j,h_i;c_{rs}).
\end{aligned} \tag{A.2}$$

For a null state in the first slot, the factorization relation is given by

$$\begin{aligned}
&\rho_a(\mathbf{L}_{-A}\chi_{rs},\mathbf{L}_{-B}\phi_2,\mathbf{L}_{-C}\phi_3)\\
&\quad=\begin{cases}\rho_a(\mathbf{L}_{-A}\phi_{h_{rs}+\frac{rs}{2}},\mathbf{L}_{-B}\phi_2,\mathbf{L}_{-C}\phi_3)\mathbf{P}^a_{rs}(h_3,h_2), & rs\in 2\mathbb{Z},\\ (-1)^{p_1+A}\rho_{1-a}(\mathbf{L}_{-A}\phi_{h_{rs}+\frac{rs}{2}},\mathbf{L}_{-B}\phi_2,\mathbf{L}_{-C}\phi_3)\mathbf{P}^a_{rs}(h_3,h_2), & rs\in 2\mathbb{Z}+1,\end{cases}
\end{aligned} \tag{A.3}$$

For a null state in the second slot,

$$\begin{aligned}
&\rho_a(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\chi_{rs},\mathbf{L}_{-E}\phi_3)\\
&\quad=\begin{cases}\mathbf{P}^a_{rs}(h_3,h_1;c_{rs})\rho_a\big(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_{h_{rs}+\frac{rs}{2}},\mathbf{L}_{-E}\phi_3\big)\,, & rs\in 2\mathbb{Z},\\ \mathbf{P}^a_{rs}(h_3,h_1;c_{rs})\rho_{1-a}\big(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_{h_{rs}+\frac{rs}{2}},\mathbf{L}_{-E}\phi_3\big)\,, & rs\in 2\mathbb{Z}+1.\end{cases}
\end{aligned} \tag{A.4}$$

and for the third slot,

$$\begin{aligned}
&\rho_a(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-E}\chi_{rs})\\
&\quad=\begin{cases}\mathbf{P}^a_{rs}(h_1,h_2;c_{rs})\rho_a\big(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-E}\phi_{h_{rs}+\frac{rs}{2}}\big)\,, & rs\in 2\mathbb{Z},\\ (-1)^{1+p_2}\mathbf{P}^a_{rs}(h_1,h_2;c_{rs})\rho_{1-a}\big(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-E}\phi_{h_{rs}+\frac{rs}{2}}\big)\,, & rs\in 2\mathbb{Z}+1.\end{cases}
\end{aligned} \tag{A.5}$$

For the case where two of the three states are null states, one have

$$\begin{aligned}\rho_f(\mathbf{L}_{-A}\chi_{rs},\mathbf{L}_{-C}\chi_{rs},\mathbf{L}_{-E}\phi_3) &= (-1)^{rs(p_1+A)}\mathbf{P}^f_{rs}(h_3,h_{rs};c_{rs})\mathbf{P}^{f+rs}_{rs}\left(h_3,h_{rs}+\frac{rs}{2};c_{rs}\right)\\ &\quad\times\rho_f(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2},\mathbf{L}_{-C}\phi_{h_{rs}+rs/2},\mathbf{L}_{-E}\phi_3),\\ \rho_f(\mathbf{L}_{-A}\chi_{rs},\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-E}\chi_{rs}) &= (-1)^{rs(1+p_1+p_2+A)}\mathbf{P}^f_{rs}(h_{rs},h_2;c_{rs})\mathbf{P}^{f+rs}_{rs}\left(h_{rs}+\frac{rs}{2},h_2;c_{rs}\right)\\ &\quad\times\rho_f(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2},\mathbf{L}_{-C}\phi_2,\mathbf{L}_{-E}\phi_{h_{rs}+rs/2}),\\ \rho_f(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\chi_{rs},\mathbf{L}_{-E}\chi_{rs}) &= (-1)^{rsp_2}\mathbf{P}^f_{rs}(h_{rs},h_1;c_{rs})\mathbf{P}^{f+rs}_{rs}\left(h_1,h_{rs}+\frac{rs}{2};c_{rs}\right)\\ &\quad\times\rho_f(\mathbf{L}_{-A}\phi_1,\mathbf{L}_{-C}\phi_{h_{rs}+rs/2},\mathbf{L}_{-E}\phi_{h_{rs}+rs/2}).\end{aligned}\tag{A.6}$$

A.2 NS null states on a (NS, R, R) sphere

The two fusion polynomial for a NS null state on a (NS, R, R) sphere is defined as

$$\begin{aligned}\mathbb{P}^{\mathrm{NS},\eta}_{rs}(\beta_2,\beta_3;c) = &\prod_{\substack{0\le k\le r-1,\ 0\le \ell\le s-1\\ k+\ell\in 2\mathbb{Z}}}\frac{2\sqrt{2}(\beta_3-\eta\beta_2)-(1-r+2k)b-(1-s+2\ell)b^{-1}}{2\sqrt{2}}\\ &\times\prod_{\substack{0\le k\le r-1,\ 0\le \ell\le s-1\\ k+\ell\in 2\mathbb{Z}+1}}\frac{2\sqrt{2}(\beta_3+\eta\beta_2)-(1-r+2k)b-(1-s+2\ell)b^{-1}}{2\sqrt{2}}.\end{aligned}\tag{A.7}$$

For the NS null state in the first slot, the factorization relation reads

$$\begin{aligned}&\rho^{(\eta)}_f(\mathbf{L}_{-A}\chi_{rs},\mathbb{L}_{-C}w^\alpha_{\beta_2},\mathbb{L}_{-E}w^\gamma_{\beta_3})\\ &\quad=\mathbb{P}^{\mathrm{NS},\eta}_{rs}(\beta_2,\beta_3;c_{rs})\begin{cases}\rho^{(\eta)}_f(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2},\mathbb{L}_{-C}w^\alpha_{\beta_2},\mathbb{L}_{-E}w^\gamma_{\beta_3}), & rs\in 2\mathbb{Z},\\ e^{i(-1)^f\pi/4}\rho^{(-\eta)}_{1-f}(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2},\mathbb{L}_{-C}w^\alpha_{\beta_2},\mathbb{L}_{-E}w^\gamma_{\beta_3}), & rs\in 2\mathbb{Z}+1.\end{cases}\end{aligned}\tag{A.8}$$

For the third slot,

$$\begin{aligned}&\rho^{(\eta)}_f(\mathbb{L}_{-A}w^\alpha_{\beta_1},\mathbb{L}_{-C}w^\gamma_{\beta_2},\mathbf{L}_{-E}\chi_{rs})\\ &\quad=\mathbb{P}^{\mathrm{NS},(-1)^f\eta}_{rs}(\beta_2,\beta_1;c_{rs})\begin{cases}\rho^{(\eta)}_f(\mathbb{L}_{-A}w^\alpha_{\beta_1},\mathbb{L}_{-C}w^\gamma_{\beta_2},\mathbf{L}_{-E}\phi_{h_{rs}+rs/2}), & rs\in 2\mathbb{Z},\\ \eta e^{i(-1)^f\pi/4}\rho^{(\eta)}_{1-f}(\mathbb{L}_{-A}w^\alpha_{\beta_1},\mathbb{L}_{-C}w^\gamma_{\beta_2},\mathbf{L}_{-E}\phi_{h_{rs}+rs/2}), & rs\in 2\mathbb{Z}+1.\end{cases}\end{aligned}\tag{A.9}$$

Finally, for the second slot with even rs, one have

$$\begin{aligned}&\rho^{(\eta)}_f(\mathbb{L}_{-A}w^\alpha_{\beta_1},\mathbf{L}_{-C}\chi_{rs},\mathbb{L}_{-E}w^\gamma_{\beta_3})\\ &\quad=\frac{1}{2}\sum_{\epsilon=\pm1}\mathbb{P}^{\mathrm{NS},\epsilon}_{rs}(\beta_3,\beta_1;c_{rs})\Big[\rho^{(\eta)}_f(\mathbb{L}_{-A}w^\alpha_{\beta_1},\mathbf{L}_{-C}\phi_{h_{rs}+rs/2},\mathbb{L}_{-E}w^\gamma_{\beta_3})\\ &\qquad\qquad-i\eta\epsilon\,\rho^{(-\eta)}_f(\mathbb{L}_{-A}w^\alpha_{\beta_1},\mathbf{L}_{-C}\phi_{h_{rs}+rs/2},\mathbb{L}_{-E}w^\gamma_{\beta_3})\Big].\end{aligned}\tag{A.10}$$

and for odd rs,

$$\begin{aligned}
&\rho_f^{(\eta)}(\mathbb{L}_{-A}w^\alpha_{\beta_1},\mathbf{L}_{-C}\chi_{rs},\mathbb{L}_{-E}w^\gamma_{\beta_3})\\
&\quad=\frac{(-1)^{(rs-1)/2}i^f e^{i\pi/4}}{2}\sum_{\epsilon=\pm1}\mathbb{P}^{\mathrm{NS},\epsilon}_{rs}(\beta_3,\beta_1;c_{rs})\Big[\eta\,\rho^{(\eta)}_{1-f}(\mathbb{L}_{-A}w^\alpha_{\beta_1},\mathbf{L}_{-C}\phi_{h_{rs}+rs/2},\mathbb{L}_{-E}w^\gamma_{\beta_3})\\
&\qquad\qquad+i\epsilon\,\rho^{(-\eta)}_{1-f}(\mathbb{L}_{-A}w^\alpha_{\beta_1},\mathbf{L}_{-C}\phi_{h_{rs}+rs/2},\mathbb{L}_{-E}w^\gamma_{\beta_3})\Big].
\end{aligned}\tag{A.11}$$

A.3 Ramond null state on a (NS, R, R) sphere

The two fusion polynomials associated with a Ramond null state are defined as

$$\begin{aligned}
\mathbb{P}^{\mathrm{R},\eta}_{rs}(h_i,\beta_j;c)=&\prod_{\substack{0\leq k\leq r-1,\ 0\leq\ell\leq s-1\\ k+\ell\in2\mathbb{Z}}}\frac{\lambda_i-2\sqrt{2}\,\eta\beta_j-(1-r+2k)b-(1-s+2\ell)b^{-1}}{2\sqrt{2}}\\
&\times\prod_{\substack{0\leq k\leq r-1,\ 0\leq\ell\leq s-1\\ k+\ell\in2\mathbb{Z}+1}}\frac{\lambda_i+2\sqrt{2}\,\eta\beta_j-(1-r+2k)b-(1-s+2\ell)b^{-1}}{2\sqrt{2}}.
\end{aligned}\tag{A.12}$$

The factorization relations are

$$\begin{aligned}
\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i,\mathbb{L}_{-C}\chi^\alpha_{rs},\mathbb{L}_{-E}w^\gamma_{\beta_j})&=\mathbb{P}^{\mathrm{R},\eta}_{rs}(h_i,\beta_j;c^{\mathrm{R}}_{rs})\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i,\mathbb{L}_{-C}w^\alpha_{\beta'_{rs}},\mathbb{L}_{-E}w^\gamma_{\beta_j}),\\
\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i,\mathbb{L}_{-C}w^\gamma_{\beta_j},\mathbb{L}_{-E}\chi^\alpha_{rs})&=\mathbb{P}^{\mathrm{R},\eta}_{rs}(h_i,\beta_j;c^{\mathrm{R}}_{rs})\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i,\mathbb{L}_{-C}w^\gamma_{\beta_j},\mathbb{L}_{-E}w^\alpha_{\beta'_{rs}}),\\
\rho_f^{(\eta)}(\mathbb{L}_{-A}\chi^\alpha_{rs},\mathbb{L}_{-C}w^\gamma_{\beta_j},\mathbf{L}_{-E}\phi_i)&=\mathbb{P}^{\mathrm{R},(-1)^f\eta}_{rs}(h_i,\beta_j;c^{\mathrm{R}}_{rs})\rho_f^{(\eta)}(\mathbb{L}_{-A}w^\alpha_{\beta'_{rs}},\mathbb{L}_{-C}w^\gamma_{\beta_j},\mathbf{L}_{-E}\phi_i),\\
\rho_f^{(\eta)}(\mathbb{L}_{-A}w^\gamma_{\beta_j},\mathbb{L}_{-C}\chi^\alpha_{rs},\mathbf{L}_{-E}\phi_i)&=\mathbb{P}^{\mathrm{R},(-1)^f\eta}_{rs}(h_i,\beta_j;c^{\mathrm{R}}_{rs})\rho_f^{(\eta)}(\mathbb{L}_{-A}w^\gamma_{\beta_j},\mathbb{L}_{-C}w^\alpha_{\beta'_{rs}},\mathbf{L}_{-E}\phi_i),
\end{aligned}\tag{A.13}$$

and for the (R, NS, R) order,

$$\begin{aligned}
&\rho_f^{(\eta)}(\mathbb{L}_{-A}\chi^\alpha_{rs},\mathbf{L}_{-C}\phi_i,\mathbb{L}_{-E}w^\gamma_{\beta_j})\\
&\quad=\frac{(-1)^{rs/2}}{2}\sum_{\epsilon=\pm1}\mathbb{P}^{\mathrm{R},\epsilon}_{rs}(h_i,\beta_j;c^{\mathrm{R}}_{rs})\begin{bmatrix}\rho_f^{(\eta)}(\mathbb{L}_{-A}w^\alpha_{\beta'_{rs}},\mathbf{L}_{-C}\phi_i,\mathbb{L}_{-E}w^\gamma_{\beta_j})\\ -i\eta\epsilon\,\rho_f^{(-\eta)}(\mathbb{L}_{-A}w^\alpha_{\beta'_{rs}},\mathbf{L}_{-C}\phi_i,\mathbb{L}_{-E}w^\gamma_{\beta_j})\end{bmatrix},\\
&\rho_f^{(\eta)}(\mathbb{L}_{-A}w^\gamma_{\beta_j},\mathbf{L}_{-C}\phi_i,\mathbb{L}_{-E}\chi^\alpha_{rs})\\
&\quad=\frac{(-1)^{rs/2}}{2}\sum_{\epsilon=\pm1}\mathbb{P}^{\mathrm{R},\epsilon}_{rs}(h_i,\beta_j;c^{\mathrm{R}}_{rs})\begin{bmatrix}\rho_f^{(\eta)}(\mathbb{L}_{-A}w^\gamma_{\beta_j},\mathbf{L}_{-C}\phi_i,\mathbb{L}_{-E}w^\alpha_{\beta'_{rs}})\\ -i\eta\epsilon\,\rho_f^{(-\eta)}(\mathbb{L}_{-A}w^\gamma_{\beta_j},\mathbf{L}_{-C}\phi_i,\mathbb{L}_{-E}w^\alpha_{\beta'_{rs}})\end{bmatrix}.
\end{aligned}\tag{A.14}$$

For the case where the two Ramond states are simutaneously null state,

$$
\begin{aligned}
&\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}\chi_{rs}^{\alpha}, \mathbb{L}_{-E}\chi_{rs}^{\gamma}) \\
&\quad = \mathbb{P}_{rs}^{\mathrm{R},\eta}(h_i, \beta_{rs}; c_{rs}^{\mathrm{R}})\mathbb{P}_{rs}^{\mathrm{R},\eta}(h_i, \beta'_{rs}; c_{rs}^{\mathrm{R}})\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}w_{\beta'_{rs}}^{\alpha}, \mathbb{L}_{-E}w_{\beta'_{rs}}^{\gamma}), \\
&\rho_f^{(\eta)}(\mathbb{L}_{-A}\chi_{rs}^{\alpha}, \mathbb{L}_{-C}\chi_{rs}^{\gamma}, \mathbf{L}_{-E}\phi_i) \\
&\quad = \mathbb{P}_{rs}^{\mathrm{R},(-1)^f\eta}(h_i, \beta_{rs}; c_{rs}^{\mathrm{R}})\mathbb{P}_{rs}^{\mathrm{R},(-1)^f\eta}(h_i, \beta'_{rs}; c_{rs}^{\mathrm{R}})\rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta'_{rs}}^{\alpha}, \mathbb{L}_{-C}w_{\beta'_{rs}}^{\gamma}, \mathbf{L}_{-E}\phi_i), \\
&\rho_f^{(\eta)}(\mathbb{L}_{-A}\chi_{rs}^{\alpha}, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}\chi_{rs}^{\gamma}) \\
&\quad = \frac{1}{2}\sum_{\epsilon=\pm1}\mathbb{P}_{rs}^{\mathrm{R},\epsilon}(h_i, \beta_{rs}; c_{rs}^{\mathrm{R}})\mathbb{P}_{rs}^{\mathrm{R},\epsilon}(h_i, \beta'_{rs}; c_{rs}^{\mathrm{R}})\begin{bmatrix}\rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta'_{rs}}^{\alpha}, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}w_{\beta'_{rs}}^{\gamma}) \\ \qquad - i\eta\epsilon\,\rho_f^{(-\eta)}(\mathbb{L}_{-A}w_{\beta'_{rs}}^{\alpha}, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}w_{\beta'_{rs}}^{\gamma})\end{bmatrix}.
\end{aligned}
\tag{A.15}
$$